

THE FRAMEWORK PROFILE C1 IMPLEMENTATION - USER MANUAL -

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7 February 2018

Revision 1.2.2
PP-UM-COR-0001

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Abstract

This document is the User Manual for the C1 Implementation of the FW Profile. The FW Profile is a specification-level modelling language defined as a restriction of UML. The core modelling constructs offered by the FW Profile are State Machines, Procedures (equivalent to UML's Activity Diagrams), and RT Containers (encapsulations of threads).

The FW Profile is implementation-independent. The C1 Implementation is a C language implementation of the modelling concepts of the FW Profile. The main features of the C1 Implementation are: small memory footprint, small CPU demands, scalability, and high reliability.

The C1 Implementation is provided with a Qualification Data Package which can be used to support the certification of applications built using its components.

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1 Introduction

This document is the User Manual for the *C1 Implementation*. The C1 Implementation is a C-language implementation of the modelling concepts of the FW Profile of reference [1]. It offers components to build State Machines, Procedures (activity diagrams) and RT Containers (encapsulations of threads). It is extensively documented and tested. An overview of the definition of the FW Profile is presented in Appendix A (State Machine Concept), in Appendix B (Procedure Concept), and in Appendix C (RT Container Concept).

The main features of the C1 Implementation are:

- **Well-Defined Semantics:** clearly and unambiguously defined behaviour.
- **Minimal Memory Requirements:** core module footprint of a few kBytes.
- **Small CPU Demands:** one single level of indirection (due to actions and guards being implemented as function pointers).
- **Scalability:** memory footprint and CPU demands are independent of number and size of state machine, procedure, and RT Container instances.
- **High Reliability:** test suite with 100% code, branch, and condition coverage (excluding error branches for system calls).
- **Formal Specification:** user requirements formally specify the implementation.
- **Requirement Traceability:** all requirements are individually traced to their implementation and to verification evidence.
- **Formal Verification:** key requirements are formally verified using the Spin verifier on a Promela model.
- **Documented Code:** doxygen documentation for all the source code.
- **Demo Application:** complete application demonstrating capabilities and mode of use.
- **Support for Extensibility:** an inheritance-like mechanism is provided through which a *derived state machine* or a *derived procedure* is created from a *base state machine* or *base procedure* by overriding some of its actions or guards.

2 Installation & Content Overview

The C1 Implementation is distributed as one single zip file. This file should be expanded in a dedicated directory. This directory becomes the *host directory* for the C1 Implementation.

The host directory contains a set of sub-directories. The sub-directories are listed in table 1. An overview of their content is presented in the following sub-sections.

The C1 Implementation software is delivered as source code and therefore no further installation operations are needed.

Table 1: Structure of Host Directory

Sub-Directory	Sub-Directory Description
doc	Holds the support documentation for the C1 Implementation (see section 2.2).
examples	Holds the source code for the coding examples (see section 2.7), including the Demo Application (see section 2.6).
log	Holds the test reports generated by the Acceptance Test Procedure (see section 2.8).
src	Holds the source code for the C1 Implementation (see section 2.3).
tests	Holds the source code for the Test Suite (see section 2.5).

2.1 Dependency on External Libraries

The State Machine and Procedure modules of the C1 Implementation only need the `stdlib` of the C language. The RT Container part needs an implementation of POSIX threads, i.e. `pthread`. If this is not available, the RT Container modules cannot be used but this has no impact on the use of the State Machine and Procedure modules.

On most linux systems, the implementation of the POSIX API is available in a library called `libpthread`.

2.2 Support Documentation

The C1 Implementation is delivered with the following support documents:

- A **FAQ Document** which answers frequently asked questions about the C1 Implementation
- The **FW Profile Definition Document** which defines the UML profile implemented by the C1 Implementation
- A **User Manual** (this document) which describes how the C1 Implementation is used
- A **User Requirement Document** which formally specifies the C1 Implementation through a set of requirements and provides validation and verification evidence for each requirement

- A **Doxygen Documentation** which provides the detailed design documentation for the C1 Implementation, its test suite and its demo application

The last three documents, together with the Test Suite, constitute the **Qualification Data Package** (QDP) for the C1 Implementation. The QDP is provided for users who need to certify their application or, more generally, who need to provide evidence of its correctness. The QDP contains the typical information which is required for software certification purposes. It can therefore be included in the certification data package of end-applications and it relieves the user of the need to produce such information for the C1 Implementation part of their applications.

2.3 Software Source Code

The source code of the C1 Implementation is stored in sub-directory `/src`. It is divided into *modules*. A module consists of a small number of C header files which define an interface to perform a set of logically related operations together with the C implementation files which implement this interface. Table 2 lists the modules in the C1 Implementation with a brief description of their content. More detailed information on the interface and implementation of each module can be found in the doxygen documentation.

The C1 Implementation modules are completely independent of each other and can be used either together or separately. Each user decides which modules to use and to link in his executable.

Table 2: Software Modules in the C1 Implementation

Mod. Name	Module Description
<i>State Machine</i>	Implementation of the State Machine Concept of the FW Profile.
<i>Procedure</i>	Implementation of the Procedure Concept of the FW Profile.
<i>RT Container</i>	Implementation of the RT Container Concept of the FW Profile.

2.4 Doxygen Documentation

The source code of the C1 Implementation, of its Test Suite and of its Demo Application is documented in accordance with doxygen rules. The entry point to the doxygen documentation is the `index.html` file in the `docs/doxygen` directory of the delivery.

2.5 Test Suite

The Test Suite is a complete application which demonstrates all aspects of the behaviour of the state machine and procedure implementations. Its implementation code is in the `tests` directory of the delivery.

The main program of the Test Suite application is in file `FwTestSuite.c`. This program consists of a set of test cases. The test cases are declared in file `FwSmTestCases.h` for the state machine part, in file `FwPrTestCases.h` for the procedure part, and in file `FwRtTestCases.h` for the RT Container part. Each test case exercises a specific aspect of the State Machine, Procedure, or RT Container behaviour.

The test cases operate on test state machines, on test procedures, and on test RT containers. The test state machines are declared in files `FwSmMakeTest.h`. The test procedures are declared in files `FwPrMakeTest.h`. The test RT containers are declared in files `FwRtMakeTest.h`.

The Test Suite offers 100% code, branch, and condition coverage of the C1 Implementation modules with the exception of the error branches in the creation and configuration operations which are entered when the application runs out of memory (i.e. when `malloc` fails) or when a POSIX system call fails.

The Test Suite application can be built by running one of the support scripts delivered with the C1 Implementation (see section 2.9).

2.6 Demo Application

The Demo Application is a complete application which demonstrates the use of the C1 Implementation by implementing a simplified but realistic monitoring system for a Hardware Device. The Demo Application is described in the Doxygen documentation of the C1 Implementation. Its implementation code is in the `examples/src/app` directory of the delivery.

The Demo Application can be built by running one of the support scripts delivered with the C1 Implementation (see section 2.9).

At present, the Demo Application does not cover the RT Container part of the C1 Implementation.

2.7 Coding Examples

Simple coding examples are provided for the procedures, the state machines, and the RT containers. Each coding example is a self-contained program which consists of a `main` program which creates, configures and runs a sample procedure, or a sample state machine, or a sample RT container. The coding example are stored in directories: `examples/src/sm_tutorials` (for the state machine part), `examples/src/pr_tutorials` (for the procedure part), and `examples/src/rt_tutorials` (for the RT container part).

The coding example programs can be built by running one of the support scripts delivered with the C1 Implementation (see section 2.9).

2.8 Acceptance Test Procedure and Test Reports

The C1 Implementation is passed through an Acceptance Test Procedure (ATP) prior to its release. The ATP is executed as a sequence of steps which are defined in table 3. For each step, a pass-fail criterium is defined. An execution of the ATP is successful if all the ATP steps satisfy their pass-fail criterium.

Table 3: Execution Steps and Pass-Fail Criteria for ATP

N	Step	Pass-Fail Criterium
1	Run Doxygen on the entire source code of the C1 Implementation, Test Suite and Demo Application	Neither errors nor warnings are reported by Doxygen
2	Compile the C1 Implementation source code files with "all warnings" enabled and with the options required to run <code>gcov</code> for both branch and statement coverage	Neither errors nor warnings are reported by the compiler
3	Compile the Test Suite source code files with "all warnings" enabled	Neither errors nor warnings are reported by the compiler
4	Build the executable to run the Test Suite for the C1 Implementation and to generate the <code>*.gcno</code> and <code>*.gcda</code> files	Neither errors nor warnings are reported by the linker
5	Run the Test Suite with Valgrind	The Test Suite runs to completion; all test cases are declared to have completed successfully; no errors are reported by Valgrind
6	Run <code>gcov</code> on all the C1 Implementation Files to which coverage requirements apply	For each C1 Implementation File to which coverage requirements apply, a <code>*.c.gcov</code> file is created and the file shows full statement and branch coverage with exception of branches entered as a result of a failure of <code>malloc</code> or of a POSIX sytem call or of branches which cannot be entered by design
7	Compile the Demo Application Files with "all warnings" enabled	Neither errors nor warnings are reported by the compiler
8	Build the executable to run the Demo Application for the C1 Implementation	Neither errors nor warnings are reported by The linker
9	Run the Demo Application with Valgrind	No errors are reported by Valgrind
10	Compile the Coding Example Files with "all warnings" enabled	Neither errors nor warnings are reported by the compiler
11	Build the executable to run the Coding Examples for the C1 Implementation	Neither errors nor warnings are reported by the linker

N	Step	Pass-Fail Criterium
12	Run the Coding Examples	No errors are reported.

The `RunAcceptanceTest.sh` shell script (not included in the delivery for end customers) automatically executes all the procedure steps described in the table and it generates several test reports. The following test reports are included in the delivery:

- Report with the code coverage information generated by `gcov`
- Report with outcome of running Valgrind on Test Suite application
- Report with outcome of running Valgrind on Demo Application

In the case of the `gcov` report, the ATP script uses `egrep` to extract all the lines of code with no statement or branch coverage together with the 6 preceding lines. This is sufficient to check whether the incomplete coverage arises in the handling of a `malloc` or POSIX system call failure or because a certain branch is impossible to enter by design (in which case a comment just before the untaken branch must be present).

2.9 Support Scripts

Two Makefiles are provided to build the executables of the C1 Implementation. These Makefile can be used with the generally available `make` tool to generate different targets. One Makefile is provided in the root and used for building the C1 implementation library and the test suite. The following targets are supported:

- `make release` Builds the fwprofile library with optimization enabled
- `make debug` Builds the fwprofile library with optimization disabled but debugging information enabled.
- `make coverage` Builds with enabled coverage
- `make coverage-info` Generates the gcov files which contain the coverage information
- `make test` Generates the test suite
- `make run-test` Runs the test suite

The second Makefile is provided in sub-directory `/examples` and is used to build the example executables. The following targets are supported:

- `make all` Builds all examples including the demo application
- `make run-all` Runs all examples including the demo application

The C1 implementation library, test suite and examples are created in the `/bin` directory.

The RT Container module requires an implementation of the POSIX library (see section 2.1). By default, the support scripts assume this library to be available in the search path under the name `libpthread`. If a different library is used, the script files must be updated. If no POSIX library is available, the script

files must be updated to remove linking of the RT Module files.

2.10 Naming Conventions

The C1 Implementation exports the following items towards users:

- Header and body files
- Global functions
- Types defined through `typedef` in `FwSmConstants.h`
- Constants and macros defined through `#define` directives

The naming conventions for these items are as follows.

The names of the header and body files of the C1 Implementation and of the global functions they export are written as a concatenation of strings (without underscores). The first letter in each string is capitalized. The names have the following form: `Fw<Xx><Name>`. The prefix `"Fw"` identifies a name as belonging to a FW Profile implementation. The string `"Xx"` identifies the domain within the FW Profile world to which the name belongs. The following values are possible for this string:

- `"Sm"` identifies a name related to the state machine domain of the FW Profile,
- `"Pr"` identifies a name related to the procedure domain of the FW Profile,
- `"Rt"` identifies a name related to the RT container domain of the FW Profile,
- `"Da"` identifies a name related to the Demo Application.

The string `"Name"` is the proper name of the function or file and it is made up of a concatenation of other strings. The following abbreviations are used in forming this name:

- `"A"`: `"action"` (as in `"action node"`)
- `"Act"`: `"action"` (as in `"action attached to a procedure node"`)
- `"Activ"`: `"activation"` (as in `"activation procedure of a RT container"`)
- `"App"`: `"application"` (as in `"the Demo Application"`)
- `"Attr"`: `"attribute"`
- `"Aux"`: `"auxiliary"` (as in `"auxiliary function"`)
- `"Config"`: `"configuration"` (as in `"the configuration of a state machine"`)
- `"Cond"`: `"condition"` (as in `"POSIX condition variable"`)
- `"Cont"`: `"container"` (as in `"RT container"`)
- `"Cnt"`: `"counter"`
- `"Cps"`: `"choice pseudo-state"`
- `"Cur"`: `"current"` (as in `"current state"` or in `"the current absorbed by a device"`)
- `"D"`: `"decision"` (as in `"decision node"`)

- "D": "dynamic" (as in "dynamic memory allocation")
- "Dec": "decision" (as in "decision node in a procedure")
- "Der": "derived" (as in "the derived state machine")
- "Desc": "descriptor" (as in "the descriptor of a state machine")
- "Err": "error" (as in "the error code field of a state machine descriptor")
- "Emb": "embedded" (as in "embedded state machine")
- "Exec": "execute" or "execution"
- "Fin": "final" (as in "final node")
- "Fps": "final pseudo-state"
- "Func": "functional" (as in "functional behaviour")
- "Ini": "initial" (as in "initial mode")
- "Init": "initialization" (as in "initialization of a state machine descriptor")
- "Ips": "initial pseudo-state"
- "Notif": "notification" (as in "notification procedure of a RT container")
- "Pr": "procedure"
- "Rec": "recursive" (as in "recursive function")
- "Rt": "real-time"
- "S": "static" (as in "static memory allocation")
- "Sm": "state machine"
- "Sta": "state" (as in "the state of a state machine")
- "Temp": "temperature" (as in "the temperature of the device")
- "Trans": "transition" (as in "the transition between two states")

The names of the types defined through `typedef` start with the string "FwSm" (in the state machine domain), or "FwPr" (in the procedure domain), or "FwRt" (in the RT container domain) and terminate with the string: "_t".

The names of the `#define` constants are written in capitals and are made up of strings concatenated with underscores.

3 State Machine Representation

This section describes how the state machine concept is implemented in the C1 Implementation. This section gives an overview description only. Detailed information is found in the Doxygen documentation. The state machine concept is described in appendix A.

3.1 State Machine Descriptor

The C1 Implementation represents a state machine through a *state machine descriptor* (SMD). A SMD is a data structure which holds all the information required to describe a state machine. Users only manipulate pointers to SMDs. These are defined as instances of type `FwSmDesc.t`. The internal structure of an SMD is described in header file `FwSmPrivate.h` and is represented in informal notation in Figure 1.

As shown in the figure, an SMD is internally split into two parts: the *base descriptor* and the *extension descriptor*. The base descriptor holds the information which defines the topology of the state machine, namely:

- The list of states in the state machine
- The list of choice pseudo-states in the state machine
- The list of transitions in the state machine

The extension descriptor holds the information which may be overridden when the state machine is extended. This consists of:

- The list of actions used in the state machine (both the state actions and the transition actions)
- The list of transition guards used in the state machine
- The list of state machines embedded in the state machine
- The pointer to the state machine data (the data upon which the state machine actions and guards operate, see Section 3.4)
- The current state of the state machine
- The previous state of the state machine
- The error code for the state machine (see Section 3.5)
- The state machine counters

Applications manipulate a state machine by passing its SMD to the functions defined by the C1 Implementation. Thus, for instance, an application executes a state machine through the following function call: `FwSmExecute(smDesc)`. Here, `smDesc` is the pointer to the SMD of the state machine to be executed.

In general, applications never need to directly access the internal fields of an SMD. They therefore do not need to be concerned with the internal structure of an SMD. Familiarity with the internal SMD structure only becomes important when, for memory or CPU efficiency reasons, users wish to by-pass some of the functions provided by the C1 Implementation and need to directly manipulate

the SMD. This is discussed further in section 6.

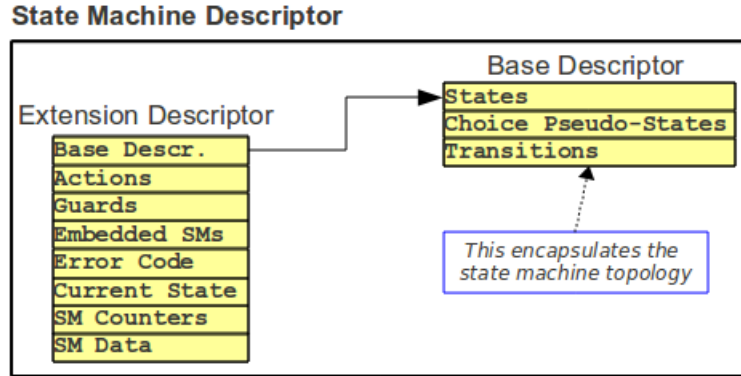


Fig. 1: Internal Structure of a State Machine Descriptor

3.2 State Machine Module

The implementation of the state machine concept is split into several files which together make up the State Machine Module (see Table 2). The files in the state machine module are listed in Table 4.

Table 4: Files in State Machine Module

Files	Description
FwSmCore.h, FwSmCore.c, FwSmPrivate.h	Provide an interface to start and stop a state machine and to send a transition command to it.
FwSmDCreate.h, FwSmDCreate.c	Provide an interface to create a new SMD. This interface is simple to use but relies on dynamic memory allocation. Applications which wish to avoid dynamic memory allocation can use the alternative services of FwSmSCreate.h.
FwSmSCreate.h, FwSmSCreate.c	Provide macros to instantiate a new SMD (without using dynamic memory allocation) and functions to initialize it. The services in this file are alternative to those of FwSmDCreate.h.
FwSmConfig.h, FwSmConfig.c	Provide an interface to configure a newly created SMD by defining its states and transitions.
FwSmAux.h, FwSmAux.c	Provides an interface to auxiliary services which are useful during the application development phase.

All applications using the C1 Implementation need the **Core** files. The **FwSmPrivate.h** header file defines the internal structure of an SMD. In most cases, applications can ignore this header file and only interact with state machines through the high-level functions declared in the other header files.

The `DCreate` and `SCreate` files are normally alternative to each other (but deployment of both in the same application is possible). Applications which are severely constrained in memory can instantiate and configure the SMDs of their state machines by directly manipulating their internal fields. This requires a detailed understanding of the internal structure of the SMD but allows an application to dispense with both the `DCreate` and `SCreate` files and with the `Config` files. An example of direct instantiation and configuration of an SMD is provided in function `FwSmMakeTestSM5Dir` in the Test Suite.

The `Aux` files are not intended for inclusion in a final application.

3.3 State Machine Actions and Guards

The state machine actions and guards are defined as function pointers of type, respectively, `FwSmAction_t` and `FwSmGuard_t`. Applications must provide functions of these two types to implement the actions and guards of their state machines. Both the guard and the action functions are called with the SMD as an argument.

Note that, if a state machine uses the same action or the same guard more than once, the associated function pointer is only stored once in the SMD.

3.4 State Machine Data

The SMD includes a field holding a pointer to the *state machine data*. The state machine data are data which are manipulated by the state machine actions and guards. The exact type of the state machine data is defined by applications for each state machine. In most cases, it will take the form a `struct` whose fields represents the inputs and outputs for the state machine actions and guards. The SMD treats the pointer to the state machine data as a pointer to `void`. Functions `FwSmSetData` and `FwSmGetData` allow this pointer to be set in and to be retrieved from an SMD.

The FW Profile allows transition commands to carry parameters and to return values. The parameters represent the parameters passed to the actions and guards triggered by the transition command and the return values represent the values returned by these actions. In the C1 Implementation a transition command is represented by an integer identifier and does not directly carry parameters or generate return values. However, the *state machine data* can be used as an equivalent mechanism through which a caller of a transition command can exchange data with the actions and guards of a state machine.

3.5 Error Checking

The state machine functions perform a limited amount of error checking. Configuration functions (namely functions in module `FwSmConfig.h`) perform consistency checks on the configuration parameters specified by the user. Details can be found in the doxygen description of the configuration functions.

The functions which trigger a transition in a state machine (namely `FwSmStart`,

FwSmMakeTrans, and **FwSmExecute**) flag the following situations as errors (both of these situations are forbidden by the FW Profile):

- A transition encounters a choice pseudo-state whose out-going transitions all have a false guard
- A transition is encountered which has a choice pseudo-state as both source and destination

Errors are reported through the *error code* field in the SMD which stores the identifier of the last error encountered by the implementation. The value of the error code can be read with the **FwSmGetErrCode** function. Nominally, the error code should be equal to **smSuccess**. If this is not the case, the behaviour of the state machine is undefined.

The error codes are listed as enumerated values in file **FwSmConstants.h**.

3.6 Else Guards

An "*Else*" guard is a guard in a transition out of a choice pseudo-state which is true when the guards of all other out-going transitions from the same choice pseudo-state are false. "Else" guards are not directly supported but their effect can be achieved as follows. If a choice pseudo-state has n outgoing transitions, where n is a natural number, their unique evaluation-order values are members of the set $\{1, \dots, n\}$ and are assigned according to the order in which they were added to the state machine when it was configured. The transitions are evaluated in increasing evaluation order. If the last transition to be added to a choice pseudo-state is given a guard which always returns true, then this transition will behave like a transition with an "*Else*" guard.

3.7 Compliance with UML State Machine Model

The definition of the state machine concept in UML is complex, often unclear, and sometimes ambiguous. The C1 Implementation adopts the state machine model of the FW Profile [1]. This is a subset of the UML model which is clearly and unambiguously defined.

4 Procedure Representation

This section describes how the procedure concept is implemented in the C1 Implementation. This section gives an overview description only. Detailed information is found in the Doxygen documentation. The procedure concept is described in appendix B.

4.1 Procedure Descriptor

The C1 Implementation represents a procedure through a *procedure descriptor* (PRD). A PRD is a data structure which holds all the information required to describe a procedure. Users only manipulate pointers to PRDs. These are defined as instances of type `FwPrDesc_t`. The internal structure of a PRD is described in header file `FwPrPrivate.h` and is represented in informal notation in Figure 2.

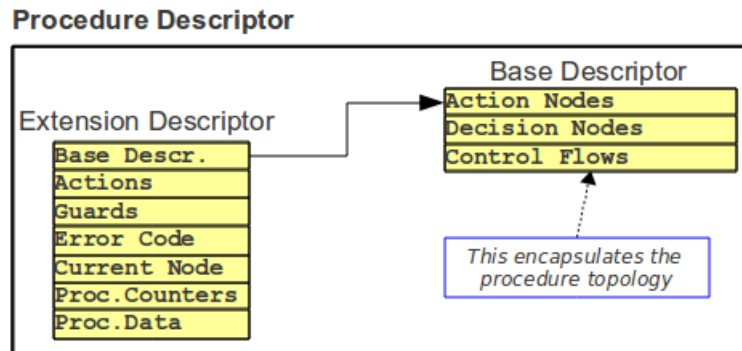


Fig. 2: Internal Structure of a Procedure Descriptor

As shown in the figure, a PRD is internally split into two parts: the *base descriptor* and the *extension descriptor*. The base descriptor holds the information which defines the topology of the procedure, namely:

- The list of action nodes in the procedure
- The list of decision nodes in the procedure
- The list of control flows in the procedure

The extension descriptor holds the information which may be overridden when the procedure is extended. This consists of:

- The list of actions used in the procedure
- The list of control flow guards used in the procedure
- The pointer to the procedure data (the data upon which the procedure actions and guards operate, see Section 4.4)
- The current node of the procedure
- The previous node of the procedure
- The error code for the procedure (see Section 4.5)

- The procedure counters

Applications manipulate a procedure by passing its PRD to the functions defined by the C1 Implementation. Thus, for instance, an application executes a procedure through the following function call: `FwPrExecute(prDesc)`. Here, `prDesc` is the pointer to the PRD of the procedure to be executed.

In general, applications never need to directly access the internal fields of a PRD. They therefore do not need to be concerned with the internal structure of a PRD. Familiarity with the internal PRD structure only becomes important when, for memory or CPU efficiency reasons, users wish to by-pass some of the functions provided by the C1 Implementation and need to directly manipulate the PRD. This is discussed further in section 7.

4.2 Procedure Module

The implementation of the procedure concept is split into several files which together make up the Procedure Module (see Table 2). The files in the state machine module are listed in Table 5.

All applications using the C1 Implementation need the `Core` module. The `FwPrPrivate.h` header file defines the internal structure of a PRD. In most cases, applications can ignore this header file and only interact with procedures through the high-level functions declared in the other header files.

The `DCreate` and `SCreate` modules are normally alternative to each other (but deployment of both in the same application is possible). Applications which are severely constrained in memory can instantiate and configure the PRDs of their procedures by directly manipulating their internal fields. This requires a detailed understanding of the internal structure of the PRD but allows an application to dispense with both the `DCreate`/`SCreate` modules and with the `Config` module. An example of direct instantiation and configuration of a PRD is provided in function `FwPrMakeTestPR2Dir` in the Test Suite.

Table 5: Files in Procedure Module

Files	Description
<code>FwPrCore.h</code> , <code>FwPrCore.c</code> , <code>FwPrPrivate.h</code>	Provides an interface to start and stop a procedure and to send a transition command to it.
<code>FwPrDCreate.h</code> , <code>FwPrDCreate.c</code>	Provides an interface to create a new PRD. This interface is simple to use but relies on dynamic memory allocation. Applications which wish to avoid dynamic memory allocation can use the alternative services of <code>FwPrSCreate.h</code> .
<code>FwPrSCreate.h</code> , <code>FwPrSCreate.c</code>	Provides macros to instantiate a new PRD (without using dynamic memory allocation) and functions to initialize it. This interface is alternative to that of <code>FwPrDCreate.h</code> .
<code>FwPrConfig.h</code> , <code>FwPrConfig.c</code>	Provides an interface to configure a newly created PRD by defining its nodes and control flows.

4.3 Procedure Actions and Guards

The procedure actions and guards are defined as function pointers of type, respectively, `FwPrAction_t` and `FwPrGuard_t`. Applications must provide functions of these two types to implement the actions and guards of their procedures. Both the guard and the action functions are called with the PRD as an argument.

Note that, if a procedure uses the same action or the same guard more than once, the associated function pointer is only stored once in the PRD.

4.4 Procedure Data

The PRD includes a field holding a pointer to the *procedure data*. The procedure data are data which are manipulated by the procedure actions and guards. The exact type of the procedure data is defined by applications for each procedure. In most cases, it will take the form a `struct` whose fields represents the inputs and outputs for the procedure actions and guards. The PRD treats the pointer to the procedure data as a pointer to `void`. Functions `FwPrSetData` and `FwPrGetData` allow this pointer to be set in and to be retrieved from a PRD.

The FW Profile allows execution commands to carry parameters and to return values. The parameters represent the parameters passed to the actions and guards triggered by the execution command and the return values represent the values returned by the actions. In the C1 Implementation, the `execute` command (function `FwPrExecute` does not directly carry parameters or generate return values. However, the *procedure data* can be used as an equivalent mechanism through which the entity which executes a procedure can exchange data with the actions and guards of a procedure.

4.5 Error Checking

The procedure functions perform a limited amount of error checking. Configuration functions (namely functions in module `FwPrConfig.h`) perform consistency checks on the configuration parameters specified by the user. Details can be found in the doxygen description of the configuration functions.

The `FwPrExecute` function which executes a procedure flags the following situation as an error: a decision node is encountered whose out-going control flows all have a false guard. Note that this situation is explicitly forbidden by the FW Profile.

Errors are reported through the *error code* field in the PRD which stores the identifier of the last error encountered by the implementation. The value of the error code can be read with the `FwPrGetErrCode` function. Nominally, the error code should be equal to `prSuccess`. If this is not the case, the behaviour of the procedure is undefined.

The error codes are listed as enumerated values in file `FwPrConstants.h`.

4.6 Else Guards

An "*Else*" guard is a guard in a control flow out of a decision node which is true when the guards of all other out-going control flows from the same decision node are false. "Else" guards are not directly supported but their effect can be achieved as follows. The out-going control flows from a decision node are evaluated in increasing evaluation order, where n is the natural number of outgoing control flows and the evaluation-order values are assigned from the set $\{1, \dots, n\}$ according to the order in which the control flows were added to the procedure when the procedure was configured. If the last control flow to be added to a decision node is given a guard which always returns true, then this transition will behave like a transition with an "*Else*" guard.

4.7 Compliance with UML Activity Diagram Model

The definition of the procedure concept in UML is complex, often unclear, and sometimes ambiguous. The C1 Implementation adopts the procedure model of the FW Profile [1]. This is a subset of the UML model which is clearly and unambiguously defined.

5 RT Container Representation

This section describes how the RT Container concept is implemented in the C1 Implementation. This section gives an overview description only. Detailed information is found in the Doxygen documentation. The RT Container concept is described in appendix C.

5.1 RT Container Descriptor

The C1 Implementation represents a RT Container through a *RT Container Descriptor* (RTD). An RTD is a data structure which holds all the information required to describe a RT Container and its current state. It is defined as an instance of type: `struct FwCrDesc`.

Users normally only manipulate pointers to RTDs. The C1 Implementation accordingly defines type `FwRtDesc_t` to represent a pointer to an RTD.

Applications manipulate a RT Container by passing its RTD to the functions defined by the C1 Implementation. Thus, for instance, an application notifies a RT Container through the following function call: `FwRtNotify(rtDesc)`. Here, `rtDesc` is the pointer to the RTD of the container to be notified (i.e. it is a variable of type `FwRtDesc_t`).

A RT Container consists of one thread (the *Activation Thread*) and two procedures (the *Activation Procedure* and the *Notification Procedure*). Within the RTD, the Activation Thread is implemented by a POSIX thread. Notification of this thread requires the use a POSIX mutex and a POSIX conditional variable. Both the mutex and the conditional variable are included in the RTD.

The Activation Procedure and Notification Procedure are represented in the RTD through the pointers to the functions implementing the procedures' actions.

Users may want to exchange data with the container procedures. For this purpose, the RTD includes a field holding a pointer to generic *container data*.

The thread and procedure information are static data which are set when the RT Container is configured and which remain constant afterwards. Additionally, the RTD also holds dynamic data which are updated during the life of the container to reflect the way it is used. The dynamic data consists of: the container state, the value of its notification counter, and the value of its error code.

Thus, in summary, the RTD holds the following data:

- A POSIX Thread to implement the Activation Thread (see section 5.2)
- A POSIX Mutex and Condition Variable to support implementation of the notification mechanism for the Activation Thread (see section 5.4)
- A set of function pointers implementing the actions of the Activation Procedure and of the Notification Procedure (see section 5.3)
- The container data (see section 5.6)

- The current state of the container (see section 5.5)
- The notification counter (see section 5.4)
- The error code for the container (see section 5.7)

The full definition of the RTD can be found in `FwRtConstants.h`.

5.2 The Activation Thread

The Activation Thread is implemented as a POSIX thread. The thread is completely encapsulated within the RT Container and users of the container do not normally need to interact with it.

The Activation Thread is created and released when the RT Container is started. By default, the thread is created with default values for all its attributes. If non-default values for the thread attributes are desired, the user can use function `FwRtSetPosixAttr` to load a POSIX thread attribute object with the desired attribute values.

5.3 RT Container Procedures

A RT Container implements the Activation Procedure and the Notification Procedure. Although these procedures are defined as standard FW Profile procedures, they are implemented within a RT Container without using the Procedure Module of the C1 Implementation. This is done in order to avoid a coupling between the RT Container Module and the Procedure Module of the C1 Implementation.

The procedure logic shown in figure 14 is therefore directly coded into the RT Container. This logic is parameterized with the functions which implement the adaptation points of the two procedures. These functions must be defined by the user when the container is configured.

The procedure functions are loaded into the container as function pointers which must conform to the `FwRtAction_t` stereotype. Functions which conform to this stereotype take the container data (see section 5.6) as a parameter and return an integer outcome. The outcome only has a meaning for functions which implement decision points for the procedures namely:

- *Implement Notification Logic* which determines whether a notification is skipped (return value is 0) or is forwarded (return value is 1)
- *Implement Activation Logic* which determines whether, in response to a notification, the container's functional behaviour is skipped (return value is 0) or is executed (return value is 1)
- *Execute Functional Behaviour* which determines whether the container's functional behaviour has terminated (return value is 1) or not (return value is 0)

In all other cases, the outcome of the procedure function is a dummy value.

For all procedure functions, default implementations are provided which do

nothing and return 1. Thus, users only need to explicitly define a procedure function when its behaviour differs from this default.

The procedure functions may only be defined at configuration time using dedicated setter functions provided by `FwRtConfig.h`. If these functions are called at other times, they will cause the container to be placed in an error state.

5.4 Notification Mechanism

Users of a RT Container send a notification to the thread encapsulated in the container by calling `FwRtNotify`. A call to this function triggers the execution of the Notification Procedure. If the notification requests is accepted by the Notification Procedure (this determined by the Implement Notification Logic action in the procedure), the value of the notification counter (variable `notifCounter` in the RTD) is incremented. The Activation Thread is released whenever the notification counter has a value greater than zero. Every release of the Activation Thread causes the counter to be decremented by 1 (see logic in section C.3).

Notification requests are therefore buffered by a RT Container. Since variable `notifCounter` is of type `FwRtCounterU2_t`, buffering will be performed up to the point where this type overflows.

The value of the Notification Counter (which is accessible through function `FwRtGetNotifCounter`) can also be used to detect overrun situations. An overrun occurs when the (n+1)-th notification is received before the Activation Procedure has completed processing of the n-th notification. This situation has arisen if, when the Activation Procedure is executed, it finds that the Notification Counter has a value greater than zero.

It is not possible to directly attach data to a notification request. Users can pass data to the notified thread through the "container data" described in section 5.6 but these data are not specific to a particular notification request. If a user needs to attach data to a notification request, it must implement a buffer within the container data structure and must implement the buffering logic in the "Implement Notification Logic" function (see figure 14).

5.5 RT Container State

The RT Container concept of appendix C recognizes two states for a RT Container: STOPPED and STARTED. The RT Container of the C1 Implementation has an expanded set of states which are shown in an informal notation in figure 3. The nominal states are:

1. `rtContUninitialized`: this is the state of the container when it is being configured, i.e. before it is initialized for the first time with function `FwRtInit` or after it has been shut down with function `FwRtShutdown`.
2. `rtContStopped`: this corresponds to state STOPPED as defined by the FW Profile. In the absence of errors, this state is entered when the container has completed its configuration and after it has been stopped with

function **FwRtStop**.

3. **rtContStarted**: this corresponds to state **STARTED** as defined by the FW Profile. In the absence of errors, this is the state of the container after it has been successfully started with function **FwRtStart** and until it is stopped with function **FwRtStop**.

Additionally, a number of error states are present. The **rtConfigErr** state is entered when a configuration function is called during normal operation (i.e. after the container has been initialized with function **FwRtInit** and before it has been shut down with function **FwRtShutdown**). An error state is also entered if a POSIX system call fails. For each kind of POSIX system call, an error state is defined. Once the container has entered an error state its behaviour is undefined. For this reason, no out-going transitions from the error states are shown in figure 3.

The range of values of the container state is defined in type **FwRtState_t**. The value of the container state can be read through function **FwRtGetContState**.

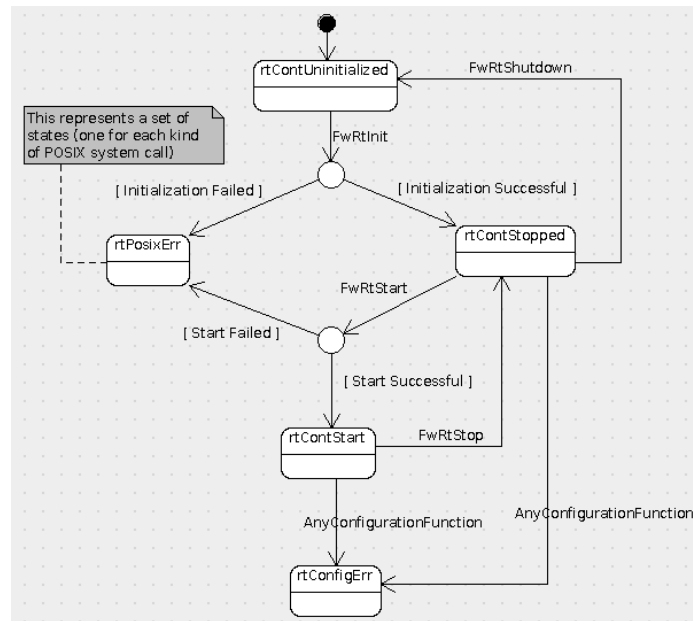


Fig. 3: RT Container States

5.6 Container Data

The RTD includes a field holding a pointer to the *container data*. The container data are data which are manipulated by the functions in the container procedures. The exact type of the container data is defined by applications for each container. In most cases, it will take the form a **struct** whose fields represent the inputs and outputs for the procedure functions. The RTD treats the pointer to the container data as a pointer to **void**. Functions **FwRtSetData** and **FwRtGetData** allow this pointer to be set in, and to be retrieved from, an RTD.

The container data may also be used as a means to attach data to a notification request (see discussion in section 5.4).

Note that some of the procedure functions may be called by (at least) two different threads: one or more external threads which notify the container by calling `FwRtNotify` and the Activation Thread which is internal to the container. If the container data are used to exchange data between these two kinds of procedure functions, then the user must implement protection mechanisms to ensure that these shared data are accessed in mutual exclusion.

5.7 Error Checking

Two forms of error checks are performed by the RT Container functions:

1. It is checked that POSIX system calls are successful. A POSIX system call fails if it returns an error code. In that case, the error code is stored in the `errCode` field of the RTD and the state of the RT Container is set to an error state which depends on the system call which reported the error (e.g. if a call to `pthread_mutex_lock` has failed, the container state is set to `rtMutexLockErr`).
2. It is checked that configuration functions are only called when the container is being configured (i.e. when it is in state `rtContUninitialized`). If a configuration function is called during normal operation, the state of the RT Container is set to the error state `rtConfigErr`.

If the RT Container is in an error state, its behaviour is undefined. The error states and the error codes are listed as enumerated values in `FwRtConstants.h`.

6 State Machine Usage

The basic mode of use of a state machine in the C1 Implementation is as follows:

- The state machine descriptor (SMD) is created
- The state machine descriptor is configured
- The state machine is sent transition commands

Examples of creation and configuration of a state machine can be found in the `FwSmMake*` functions of the `FwSmMakeTest.h` module in the Test Suite. Examples of state machine commanding can be found in the `FwSmTestCases.h` module in the Test Suite and in the Demo Application.

The pseudo-code examples in this section refer to the test state machine SM5 which is shown in Figure 4. This test state machine is built by function `FwSmMakeTestSM5` in the Test Suite and it is used in a number of test cases in the test suite.

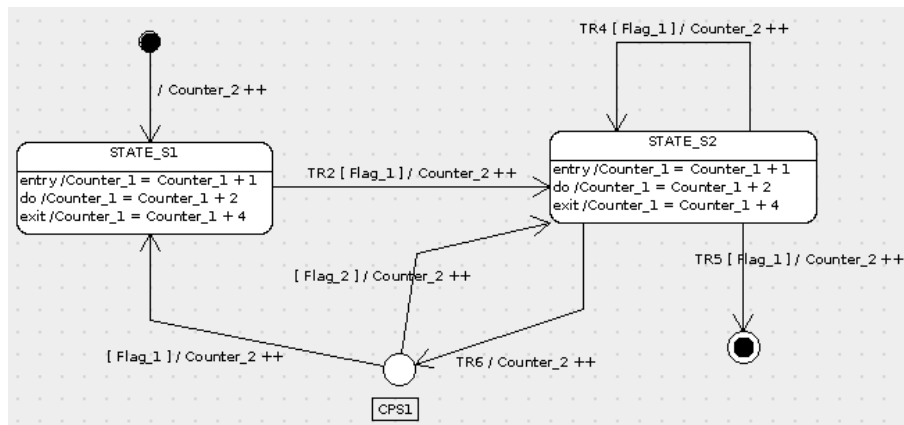


Fig. 4: Test State Machine SM5

6.1 State Machine Descriptor Creation

In the state machine creation process, a state machine descriptor together with all its internal data structures is instantiated and initialized.

A state machine descriptor can be created in one of three alternative ways as described in Table 6. The last column in the table gives a pointer to one or more functions in the Test Suite where each creation method is demonstrated.

Table 6: Methods to Create a State Machine

Method	Description	SM Example
<i>Dynamic Creation</i>	Creation is done through the FwSmDCreate function. The caller specifies the size of the state machine and the function allocates the memory for the SMD and its internal data structures and returns a pointer to the SMD. This creation interface is simple but relies on dynamic memory allocation (malloc). Release of the memory allocated at creation can be done with functions FwSmRelease and FwSmReleaseRec .	MakeTestSM1 , MakeTestSM5
<i>Static Creation</i>	Creation is done in two steps. First, the SMD and its internal data structures are instantiated using macro FW_SM_INST (if the state machine has choice pseudo-states) or macro FW_SM_INST_NOCP (if the state machine has no choice pseudo-states), and then the SMD and its internal data structures are initialized using the FwSmInit function. No dynamic memory allocation is used.	MakeTestSM1Static , MakeTestSM5Static
<i>Direct Creation</i>	The application directly instantiates the internal data structures of the state machine descriptor. Memory footprint is reduced because neither the FwSmDCreate function nor the FwSmInit is needed but users must understand the internal structure of an SMD (this is defined in header file FwSmPrivate.h).	MakeTestSM5Dir

The listings below illustrate the three ways to create an SMD for the case of the test state machine SM5 (see Figure 4). The characteristics of the state machine are: 2 states, 1 choice pseudo-state, 7 transitions, 4 actions and 2 guards. With reference to the number of actions and of guards, it is recalled that actions which appear more than once are counted only once (in state machine SM5, the state actions appear twice because the two states have the same actions and the transition action appears seven times because all transitions have the same transition action). Similarly, guards which appear more than once are counted only once (in state machine SM5, the guard `Flag_1` occurs four times and the guard `Flag_2` occurs once).

Note that, in listing 1 (dynamic creation case), the variable `smDesc` holds a pointer to the SMD whereas in listings 2 and 3 (static and direct creation), it holds the SMD itself. Most users should use either the approach of listing 1 or that of listing 2. The approach of listing 3 requires a detailed understanding of the internal organization of an SMD and should only be used in applications where memory requirements are so tight that it is desirable to drop the SMD creation functions provided by the C1 Implementation.

```
1  /* Create and initialize the state machine descriptor */
2  FwSmDesc_t smDesc = FwSmCreate(2, 1, 7, 4, 2);
```

Listing 1: Dynamic Creation of Test State Machine SM5

```
1  /* Instantiate data structures for state machine descriptor */
2  FW_SM_INST(smDesc, 2, 1, 7, 4, 2)
3
4  /* Initialize data structures for state machine descriptor */
5  FwSmInit(&smDesc);
```

Listing 2: Static Creation of Test State Machine SM5

```
1  /* Instantiate data structures for state machine descriptor */
2  static SmTrans_t trans[7];
3  static FwSmAction_t actions[5];
4  static FwSmGuard_t guards[3];
5  static SmPState_t pStates[2];
6  static SmCState_t cStates[1];
7  static FwSmDesc_t esmDesc[2];
8  static struct FwSmDesc smDesc;
9  static SmBaseDesc_t smBase;
10
11 /* Initialize state machine descriptor */
12 smBase.pStates = pStates;
13 smBase.cStates = cStates;
14 smBase.trans = trans;
15 smBase.nOfPStates = 2;
16 smBase.nOfCStates = 1;
17 smBase.nOfTrans = 7;
18 smDesc.smBase = &smBase;
19 smDesc.transCnt = 0;
20 smDesc.curState = 0;
21 smDesc.prevState = 0;
22 smDesc.smData = NULL;
23 smDesc.nOfActions = 5;
24 smDesc.nOfGuards = 3;
```

```

25 smDesc.smActions = actions;
26 smDesc.smGuards = guards;
27 smDesc.esmDesc = esmDesc;
28 smDesc.errCode = success;
29 smDesc.smData = NULL;

```

Listing 3: Direct Creation of Test State Machine SM5

6.2 State Machine Descriptor Configuration

After being created, an SMD is initialized but is not yet configured. Configuration is done using the functions defined in header file `FwSmConfig.h`. Configuration is done in steps as follows:

1. The states of the state machine are defined with the `FwSmAddState` function.
2. The choice pseudo-states of the state machine are defined with the `FwSmAddChoicePseudoState` function.
3. The transitions of the state machines are defined with the `FwSmAddTrans*` functions (there are several of these functions, one for each type of transition source and destination).
4. The pointer to the state machine data in the state machine descriptor is set with the `FwSmSetData` function.
5. The consistency and completeness of the state machine configuration may be verified with function `FwSmCheck` or `FwSmCheckRec`.
6. The configuration of a state machine can be printed with function `FwSmPrintConfig`.

The only constraint on the order in which these steps are performed is that a transition from a state or choice pseudo-state can only be defined after the source state or choice pseudo-state has been defined.

Configuration of a state machine can only be done once: a state machine which has already been configured cannot be configured again.

The pseudo-code in listing 4 shows configuration steps 1 to 3 for the case of the test state machine SM5 (see Figure 4). In the pseudo-code, the variables with names like `incrCnt1By2` ("increment counter 1 by 2") are the functions representing the actions in the state machine and the variables with names like `retFlag1` ("return Flag_1") are the functions representing the guards in the state machine. The variables `CPS1`, `STATE_S1` and `STATE_S2` are the identifiers of the choice pseudo-state and of the two states of the state machine. The variables `TR2` to `TR6` are the identifiers of the transition commands in the state machine.

Note that the identifiers of choice pseudo-states and states must be integers in the range 1 to *n* where *n* is, respectively, the number of choice pseudo-states or the number of states in the state machine. The identifiers of the transition commands are non-negative integers but they are not constrained to be sequential.

Users who are severely constrained in memory can avoid linking the configuration functions by directly configuring an SMD and its internal data structures. This, however, requires an understanding of the internal structure of an SMD (this is defined in header file `FwSmPrivate.h`). An example of direct SMD configuration can be found in function `FwSmMakeTestSM5Dir` in the Test Suite. The memory saving of taking this approach is about 2 kBytes (this is the memory footprint of the `FwSmConfig.h` module, see Section 9.2). The pseudo-code of listing 5 re-casts the pseudo-code of listing 4 to perform a direct configuration of test state machine SM5 (see Figure 4).

```

1  /* Configure the states */
2  FwSmAddState(smDesc, STATE_S1, 1, &incrCnt1By1, &incrCnt1By4, &
   incrCnt1By2, NULL);
3  FwSmAddState(smDesc, STATE_S2, 3, &incrCnt1By1, &incrCnt1By4, &
   incrCnt1By2, NULL);
4
5  /* Configure the choice pseudo-state */
6  FwSmAddChoicePseudoState(smDesc, CPS1, 2);
7
8  /* Configure the state transitions */
9  FwSmAddTransIpsToSta(smDesc, STATE_S1, &incrCnt2By1);
10 FwSmAddTransStaToSta(smDesc, TR2, STATE_S1, STATE_S2, &incrCnt2By1,
   &retFlag1);
11 FwSmAddTransStaToCps(smDesc, TR6, STATE_S2, CPS1, &incrCnt2By1,
   NULL);
12 FwSmAddTransCpsToSta(smDesc, CPS1, STATE_S1, &incrCnt2By1, &
   retFlag1);
13 FwSmAddTransCpsToSta(smDesc, CPS1, STATE_S2, &incrCnt2By1, &
   retFlag2);
14 FwSmAddTransStaToFps(smDesc, TR5, STATE_S2, &incrCnt2By1, &retFlag1
   );
15 FwSmAddTransStaToSta(smDesc, TR4, STATE_S2, STATE_S2, &incrCnt2By1,
   &retFlag1);

```

Listing 4: Configuration of Test State Machine SM5

```

1  /* Configure the array of state machine actions */
2  actions[0] = &SmDummyAction;
3  actions[1] = &incrCnt1By1;
4  actions[2] = &incrCnt1By2;
5  actions[3] = &incrCnt1By4;
6  actions[4] = &incrCnt2By1;
7
8  /* Configure the array of state machine guards */
9  guards[0] = &SmDummyGuard;
10 guards[1] = &retFlag1;
11 guards[2] = &retFlag2;
12
13 /* Configure the array of embedded state machines */
14 esmDesc[0] = NULL;
15 esmDesc[1] = NULL;
16
17 /* Configure the array of proper states */
18 pStates[0].outTransIndex = 1;
19 pStates[0].nOfOutTrans = 1;
20 pStates[0].iEntryAction = 1;
21 pStates[0].iDoAction = 2;
22 pStates[0].iExitAction = 3;
23

```

```

24 pStates[1].outTransIndex = 2;
25 pStates[1].nOfOutTrans = 3;
26 pStates[1].iEntryAction = 1;
27 pStates[1].iDoAction = 2;
28 pStates[1].iExitAction = 3;
29
30 /* Configure the array of choice pseudo-states */
31 cStates[0].outTransIndex = 5;
32 cStates[0].nOfOutTrans = 2;
33
34 /* Configure the array of transitions */
35 trans[0].dest = STATE_S1;
36 trans[0].id = 0;
37 trans[0].iTrAction = 4;
38 trans[0].iTrGuard = 0;
39
40 trans[1].dest = STATE_S2;
41 trans[1].id = TR2;
42 trans[1].iTrAction = 4;
43 trans[1].iTrGuard = 1;
44
45 trans[2].dest = -CPS1;
46 trans[2].id = TR6;
47 trans[2].iTrAction = 4;
48 trans[2].iTrGuard = 0;
49
50 trans[3].dest = 0;
51 trans[3].id = TR5;
52 trans[3].iTrAction = 4;
53 trans[3].iTrGuard = 1;
54
55 trans[4].dest = STATE_S2;
56 trans[4].id = TR4;
57 trans[4].iTrAction = 4;
58 trans[4].iTrGuard = 1;
59
60 trans[5].dest = STATE_S1;
61 trans[5].id = -1;
62 trans[5].iTrAction = 4;
63 trans[5].iTrGuard = 1;
64
65 trans[6].dest = STATE_S2;
66 trans[6].id = -1;
67 trans[6].iTrAction = 4;
68 trans[6].iTrGuard = 2;

```

Listing 5: Direct Configuration of Test State Machine SM5

6.3 State Machine Execution

A state machine is executed by sending it a command to perform a transition. In accordance with the FW Profile, before being executed, a state machine must be started. This is done with the `FwSmStart` function. Applications can check whether a state machine has already been started with function `FwSmIsStarted`.

After a state machine has been started, it will respond to requests to perform a transition (but note that sending such a request to a state machine which has not been started is not an error - the request is simply ignored). State transitions are exclusively triggered by *transition commands*. Each state machine reacts

to a finite number of transition commands. Each transition command has an identifier (a non-negative integer).

The previous state is set to an invalid value when the state machine is started. Before the current state is changed, the current state is copied to the previous state. Thus, the previous state remains invalid when the initial pseudo-state is left, and a transition to the same state makes the previous and current states equal. Stopping the state machine sets both states to invalid values. If several transitions with the same trigger are available, their guards are evaluated in the order in which the transitions were added to the state machine; the first transition with a true guard is selected.

The range of transition command identifiers is defined by a user when a state machine is configured. The identifier 0 is reserved for the *Execute* transition command. The Execute transition command is a transition command which triggers the execution of the do-action of the current state of a state machine. The `#define` constant `FW_TR_EXECUTE` is provided to represent the identifier of the "Execute" transition.

A transition command is sent to a state machine with function `FwSmMakeTrans`. As a matter of convenience, function `FwSmExecute` is provided to send the Execute command to a state machine.

A state machine is stopped with function `FwSmStop`. State machines can be started and stopped as many times as desired.

The example in listing 6 shows a short commanding sequence for the SM5 state machine. In the pseudo-code, variable `smDesc` holds the pointer to the SMD (i.e. variable `smDesc` is of type `FwSmDesc_t`). The state machine is first started. This brings it to state `S1`. The state machine is then sent transition commands `TR2` and `TR6`. The response of the state machine to these commands depends on the values of the flags `Flag_1` and `Flag_2`. If, for instance, `Flag_1` is true and `Flag_2` is false, then the first transition command brings the state machine to `S2` and the second one brings it back to `S1`.

```

1  /* Start the state machine */
2  FwSmStart(smDesc);
3
4  /* Send transition command TR2 to the state machine */
5  FwSmMakeTrans(smDesc, TR2);
6
7  /* Send transition command TR6 to the state machine */
8  FwSmMakeTrans(smDesc, TR6);
9
10 /* Stop the state machine */
11 FwSmStop(smDesc);

```

Listing 6: Commanding Sequence for Test State Machine SM5

6.4 State Machine Extension

The C1 Implementation supports an extension mechanism for state machines which is similar to the inheritance-based extension mechanism of object-oriented

languages. The extension mechanism is *optional*: there is no requirement that all applications using the C1 Implementation also use the extension mechanism.

A state machine (the *base state machine*) can be *extended* to create a new state machine (the *derived state machine*). A derived state machine can either be created dynamically with the `FwSmDCreateDer` function or else it can be instantiated statically with macro `FW_SM_INST_DER` and initialized with function `FwSmInitDer`.

After being created, a derived state machine is a clone of its base. It can then be configured by performing one or more of the following operations:

- Overriding its actions (through function `FwSmOverrideAction`)
- Overriding its guards (through function `FwSmOverrideGuard`)
- Embedding new state machines in its states (through function `FwSmEmbed`).

The internal structure of the SMD is designed to minimize the memory requirements of derived state machines (see Figure 1). An SMD is split into two parts: the Base Descriptor and the Extension Descriptor. The Base Descriptor holds the information about the state machine topology (its states, choice pseudo-states and their connections) whereas the Extension Descriptor holds the information about the state machine actions and guards and its embedded state machines. During the extension process, only the Extension Descriptor is duplicated whereas the Base Descriptor is shared between a state machine and its children (see Figure 5). This significantly reduces memory occupation in a situation where a large number of state machines are derived from the same base state machine.

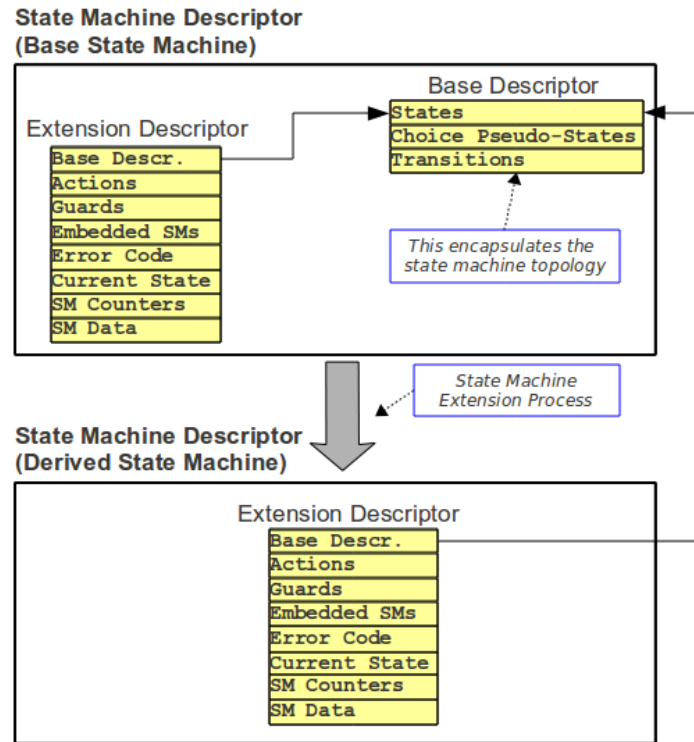


Fig. 5: Extension Mechanism for State Machine Descriptors

The extension mechanism is useful where there is a need to define a large number of state machines which share the same topology (same set of states, of choice pseudo-states, and of transitions) but differ either in their actions, or in their guards, or in the internal behaviour of their states.

As an example, consider an application which manages a set of external hardware devices all of which are characterized by the same basic states (e.g. OFF, STANDBY, OPERATIONAL) and by the same behaviour in states OFF and OPERATIONAL, but which have different and device-specific behaviour in state STANDBY. In this case, it is convenient to proceed as follows:

- A base state machine is defined to model the behaviour which is shared by all devices
- For each device, a state machine is derived which overrides the behaviour in state STANDBY in a manner that is specific to each device.

The Demo Application offers another example of a situation where the extension mechanism is useful. In this application, several Failure Detection (FD) Checks must be implemented for the same Hardware Device. All FD Checks share the same basic behaviour: they can be enabled and disabled and, when they are enabled, they can either declare the device to be healthy or they can declare it to have failed. The algorithm which is used to declare the device healthy or failed is, however, specific to each FD Check. The application is therefore organized as follows:

- A base state machine is defined to model the behaviour which is shared by all FD Checks
- For each FD Check, a state machine is derived which overrides the algorithm to check the health of the device

The pseudo-code in listing 7 offers a concrete example of creation and configuration of a derived state machine. The state machine of Figure 4 acts as *base state machine* and it is extended to create a new state machine (the *derived state machine*) as shown in Figure 5. The derived state machine has overridden the entry action of the two states and the guard on the transition from the choice pseudo-state to state S2. Note that it would not have been possible to override only the entry action of state S1. The entry actions of the two states S1 and S2 have been defined to be identical and are implemented by the same function `incrCnt1By1` (see configuration examples in the previous sections) and can therefore only be overridden together.

In the example of listing 7, the descriptor of the derived state machine is created dynamically by function `FwSmCreateDer`. For users who do not wish to rely on dynamic memory allocation, an alternative approach is available which is illustrated in listing 8. Note that in listing 7 the first example variable `smDescDer` is a pointer to the state machine descriptor whereas in listing 8 it is the state machine descriptor itself.

Use of the derived state machine is done as in the case of non-derived state machines and the examples of section 6.3 remain therefore applicable.

```

1  /* Create the derived state machine (smDesc is the pointer to the
2     SMD of the base SM) */
3  FwSmDesc_t smDescDer = FwSmCreateDer(smDesc);
4
5  /* Override Action incrCnt1By1 with action incrCnt1By8 */
6  FwSmOverrideAction(smDescDer, &incrCnt1By1, &incrCnt1By8);
7
8  /* Override Guard retFlag1 with guard retFlag3 */
9  FwSmOverrideGuard(smDescDer, &retFlag1, &retFlag3);

```

Listing 7: Dynamic Creation and Configuration of Derived State Machine

```

1  /* Instantiate derived SM with 2 states, 4 actions and 2 guards */
2  FW_SM_INST_DER(smDescDer, 2, 4, 2)
3
4  /* Create the derived state machine (smDesc is the pointer to the
5     SMD of the base SM) */
6  FwSmInitDer(&smDescDer, smDesc);
7
8  /* Override Action incrCnt1By1 with action incrCnt1By8 */
9  FwSmOverrideAction(&smDescDer, &incrCnt1By1, &incrCnt1By8);
10
11 /* Override Guard retFlag1 with guard retFlag3 */
12 FwSmOverrideGuard(&smDescDer, &retFlag1, &retFlag3);

```

Listing 8: Static Creation and Configuration of Derived State Machine

7 Procedure Usage

The basic mode of use of a procedure in the C1 Implementation is as follows:

- The procedure descriptor (PRD) is created
- The procedure descriptor is configured
- The procedure is executed

Examples of creation and configuration of a procedure can be found in the **FwPrMake*** functions of the **FwPrMakeTest.h** module in the Test Suite. Examples of procedure execution can be found in the **FwPrTestCases.c** module in the Test Suite and in the Demo Application.

The pseudo-code examples in this section refer to the test procedure PR2 which is shown in Figure 6. This test procedure is built by function **FwPrMakeTestPR2** in the Test Suite and it is used in a number of test cases in the test suite.

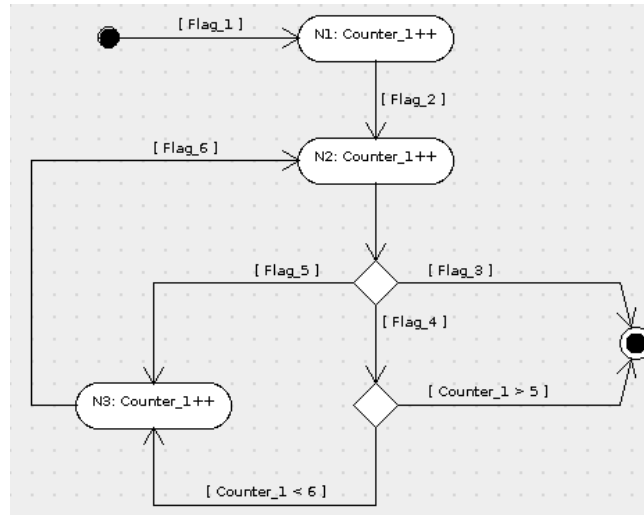


Fig. 6: Test procedure PR2

7.1 Procedure Descriptor Creation

In the procedure creation process, a procedure descriptor together with all its internal data structures is instantiated and initialized.

A procedure descriptor can be created in one of three alternative ways as described in Table 7. The last column in the table gives a pointer to one or more functions in the Test Suite where each creation method is demonstrated.

Table 7: Methods to Create a Procedure

Method	Description	Proc. Example
<i>Dynamic Creation</i>	Creation is done through the FwPrDCreate function. The caller specifies the size of the procedure and the function allocates the memory for the PRD and its internal data structures and returns a pointer to the PRD. This creation interface is simple but requires dynamic memory allocation (malloc). Release of the memory allocated at creation can be done with function FwPrRelease .	MakeTestPR1 , MakeTestPR2
<i>Static Creation</i>	Creation is done in two steps. First, the PRD and its internal data structures are instantiated using either macro FW_PR_INST (if the procedure has at least one decision node) or macro FW_PR_INST_NODEC (if the procedure has no decision nodes), and then the PRD and its internal data structures are initialized using the FwPrInit function. No dynamic memory allocation is used.	MakeTestPR1Static , MakeTestPR2Static
<i>Direct Creation</i>	The application directly instantiates the internal data structures of the procedure descriptor. Memory footprint is reduced because neither the FwPrDCreate function nor the FwPrInit function is needed but users must understand the internal structure of a PRD (this is defined in header file FwPrPrivate.h).	MakeTestPR2Dir

The examples of listing 9 to 11 illustrate the three ways to create a PRD for the case of the test procedure PR2 (see Figure 6). The characteristics of the procedure are: 3 action nodes, 2 decision nodes, 9 control flows, 1 action (used in several actions nodes), and 8 guards. With reference to the number of actions and of guards, it is recalled that actions and guards which appear more than once are counted only once (in procedure PR2, the action to increment `Counter_1` by 1 occurs three times). Note that, in listing 9 (dynamic creation case), the variable `prDesc` holds a pointer to the PRD whereas in listings 10 and 11 (static and direct creation), it holds the PRD itself.

Most users should use either the approach of listing 9 or that of listing 10. The approach of listing 11 requires a detailed understanding of the internal organization of a PRD and should only be used in applications where memory requirements are so tight that it is desirable to drop the PRD creation functions provided by the C1 Implementation.

```
1 /* Create and initialize the procedure descriptor */
2 FwPrDesc_t prDesc = FwPrCreate(3, 2, 9, 1, 8);
```

Listing 9: Dynamic Creation of Test Procedure PR2

```
1 /* Instantiate data structures for procedure descriptor */
2 FW_PR_INST(prDesc, 3, 2, 9, 1, 8)
3
4 /* Initialize data structures for procedure descriptor */
5 FwPrInit(&prDesc);
```

Listing 10: Static Creation of Test Procedure PR2

```
1 /* Instantiate data structures for procedure descriptor */
2 static PrANode_t aNodes[3];
3 static PrDNode_t dNodes[2];
4 static PrFlow_t flows[9];
5 static FwPrAction_t actions[1];
6 static FwPrGuard_t guards[9];
7 static PrBaseDesc_t prBase;
8 static struct FwPrDesc prDesc;
9
10 /* Initialize procedure descriptor */
11 prBase.aNodes = aNodes;
12 prBase.dNodes = dNodes;
13 prBase.flows = flows;
14 prBase.nOfANodes = 3;
15 prBase.nOfDNodes = 2;
16 prBase.nOfFlows = 9;
17 prDesc.curNode = 0;
18 prDesc.prevNode = 0;
19 prDesc.errCode = prSuccess;
20 prDesc.flowCnt = 0;
21 prDesc.nOfActions = 1;
22 prDesc.nOfGuards = 9;
23 prDesc.prActions = actions;
24 prDesc.prBase = prBase;
25 prDesc.prData = prData;
26 prDesc.prGuards = guards;
27 prDesc.prData = NULL;
```

Listing 11: Direct Creation of Test Procedure PR2

7.2 Procedure Descriptor Configuration

After being created, a PRD is initialized but is not yet configured. Configuration is done using the functions defined in header file `FwPrConfig.h`. Configuration is done in steps as follows:

1. The action nodes of the procedure are defined with the `FwPrAddActionNode` function.
2. The decision nodes of the procedure are defined with the `FwPrAddDecisionNode` function.
3. The control flows of the procedures are defined with the `FwPrAddFlow*` functions (there are several of these functions, one for each type of control flow source and destination).
4. The pointer to the procedure data in the procedure descriptor is set with the `FwPrSetData` function.
5. The consistency and completeness of the procedure configuration may be verified with function `FwPrCheck`.

The only constraint on the order in which these steps are performed is that a control flow can only be defined after its source node has been defined.

Configuration of a procedure can only be done once: a procedure which has already been configured cannot be configured again.

Listing 12 shows configuration steps 1 to 3 for the case of the test procedure PR2 (see Figure 6). In the pseudo-code, the variable `incrCnt1By1` ("increment counter 1 by 1") is the function representing the actions in the procedure and the variables with names like `retFlag1` ("return Flag_1") are the functions representing the guards in the procedure.

Users who are severely constrained in memory can avoid linking the configuration functions by directly configuring a PRD and its internal data structures. This, however, requires an understanding of the internal structure of a PRD (this is defined in header file `FwPrPrivate.h`). An example of direct PRD configuration can be found in function `FwPrMakeTestPR2Dir` in the Test Suite. The memory saving of taking this approach is about 2 kBytes (this is the memory footprint of the `FwPrConfig.h` module, see Section 9.2). Listing 13 re-casts the example of listing 12 to perform a direct configuration of test procedure PR2 (see Figure 6).

```

1  /* Configure the action nodes */
2  FwPrAddActionNode(prDesc, N1, &incrCnt1By1);
3  FwPrAddActionNode(prDesc, N2, &incrCnt1By1);
4  FwPrAddActionNode(prDesc, N3, &incrCnt1By1);
5
6  /* Configure the decision nodes */
7  FwPrAddDecisionNode(prDesc, D1, 3);
8  FwPrAddDecisionNode(prDesc, D2, 2);

```

```

9
10 /* Configure the control flows */
11 FwPrAddFlowIniToAct(prDesc, N1, &retFlag1);
12 FwPrAddFlowActToAct(prDesc, N1, N2, &retFlag2);
13 FwPrAddFlowActToDec(prDesc, N2, D1, NULL);
14 FwPrAddFlowDecToFin(prDesc, D1, &retFlag3);
15 FwPrAddFlowDecToDec(prDesc, D1, D2, &retFlag4);
16 FwPrAddFlowDecToAct(prDesc, D1, N3, &retFlag5);
17 FwPrAddFlowActToAct(prDesc, N3, N2, &retFlag6);
18 FwPrAddFlowDecToAct(prDesc, D2, N3, &returnCounter1SmallerThan6);
19 FwPrAddFlowDecToFin(prDesc, D2, &returnCounter1GreaterThan5);

```

Listing 12: Configuration of Test Procedure PR2

```

1 /* Configure the array of procedure actions */
2 actions[0] = &incrCnt1By1;
3
4 /* Configure the array of procedure guards */
5 guards[0] = &PrDummyGuard;
6 guards[1] = &retFlag1;
7 guards[2] = &retFlag2;
8 guards[3] = &retFlag3;
9 guards[4] = &retFlag4;
10 guards[5] = &retFlag5;
11 guards[6] = &retFlag6;
12 guards[7] = &returnCounter1GreaterThan5;
13 guards[8] = &returnCounter1SmallerThan6;
14
15 /* Configure the array of action nodes */
16 aNodes[0].iAction = 0;
17 aNodes[0].iFlow = 1;
18 aNodes[1].iAction = 0;
19 aNodes[1].iFlow = 2;
20 aNodes[2].iAction = 0;
21 aNodes[2].iFlow = 3;
22
23 /* Configure the array of decision nodes */
24 dNodes[0].nOfOutTrans = 3;
25 dNodes[0].outFlowIndex = 4;
26 dNodes[1].nOfOutTrans = 2;
27 dNodes[1].outFlowIndex = 7;
28
29 /* Configure the array of control flows */
30 flows[0].dest = 1;
31 flows[0].iGuard = 1;
32 flows[1].dest = 2;
33 flows[1].iGuard = 2;
34 flows[2].dest = -1;
35 flows[2].iGuard = 0;
36 flows[3].dest = 2;
37 flows[3].iGuard = 6;
38 flows[4].dest = 3;
39 flows[4].iGuard = 5;
40 flows[5].dest = -2;
41 flows[5].iGuard = 4;
42 flows[6].dest = 0;
43 flows[6].iGuard = 3;
44 flows[7].dest = 3;
45 flows[7].iGuard = 8;
46 flows[8].dest = 0;
47 flows[8].iGuard = 7;

```

Listing 13: Direct Configuration of Test Procedure PR2

7.3 Procedure Execution

In accordance with the FW Profile, before being executed, a procedure must be started. This is done with the **FwPrStart** function. Applications can check whether a procedure has already been started with function **FwPrIsStarted**.

After a procedure has been started, it will respond to execution requests (but note that sending such a request to a procedure which has not been started is not an error - the execution request is simply ignored). An execution request is sent to a procedure by means of function **FwPrExecute**.

The previous node is set to an invalid value when the procedure is started. When execution enters an action node, the current node is copied to the previous node before the current node is changed. Thus, the previous node remains invalid when the initial node is left, and a self-transition makes the previous and current nodes equal. Stopping the procedure sets both nodes to invalid values. If a decision node with n outgoing control flows is encountered, where n is a natural number, each control flow is assigned an evaluation-order value from the set $\{1, \dots, n\}$ according to the order in which it was added to the procedure. The guards are evaluated in increasing evaluation order, and the control flow with the lowest evaluation-order value whose guard evaluates to TRUE is selected.

A procedure is stopped with function **FwPrStop**. Procedures can be started and stopped as many times as desired.

Listing 14 shows a short commanding sequence for the PR2 procedure. In the pseudo-code, variable **prDesc** holds the pointer to the PRD (i.e. variable **prDesc** is of type **FwPrDesc_t**). The procedure is first started. It is then executed twice and it is finally stopped. The response of the procedure to these commands depends on the values of the flags **Flag_1** and **Flag_2**. If, for instance, **Flag_1** is true and **Flag_2** is false, then the first execution command brings the procedure to N1 (causing its action to be executed) and the second one has no effect.

```

1  /* Start the procedure */
2  FwPrStart(prDesc);
3
4  /* Send first execution command to the procedure */
5  FwPrExecute(prDesc);
6
7  /* Send second execution command to the procedure */
8  FwPrExecute(prDesc);
9
10 /* Stop the procedure */
11 FwPrStop(prDesc);

```

Listing 14: Commanding Sequence for Test Procedure PR2

7.4 Procedure Extension

The C1 Implementation supports an extension mechanism for procedures which is similar to the inheritance-based extension mechanism of object-oriented languages. The extension mechanism is *optional*: there is no requirement that all applications using the C1 Implementation also use the extension mechanism.

A procedure (the *base procedure*) can be *extended* to create a new procedure (the *derived procedure*). A derived procedure can either be created dynamically with the `FwPrDCreateDer` function or else it can be instantiated statically with macro `FW_PR_INST_DER` and initialized with function `FwPrInitDer`.

After being created, a derived procedure is a clone of its base. It can then be configured by performing one or more of the following operations:

- Overriding its actions (through function `FwPrOverrideAction`)
- Overriding its guards (through function `FwPrOverrideGuard`)

The internal structure of the PRD is designed to minimize the memory requirements of derived procedures (see Figure 2). A PRD is split into two parts: the Base Descriptor and the Extension Descriptor. The Base Descriptor holds the information about the procedure topology (its nodes and the control flows connecting them) whereas the Extension Descriptor holds the information about the procedure actions and guards and the procedure state. During the extension process, only the Extension Descriptor is duplicated whereas the Base Descriptor is shared between a procedure and its children (see Figure 7). This significantly reduces memory occupation in a situation where a large number of procedures are derived from the same base procedure.

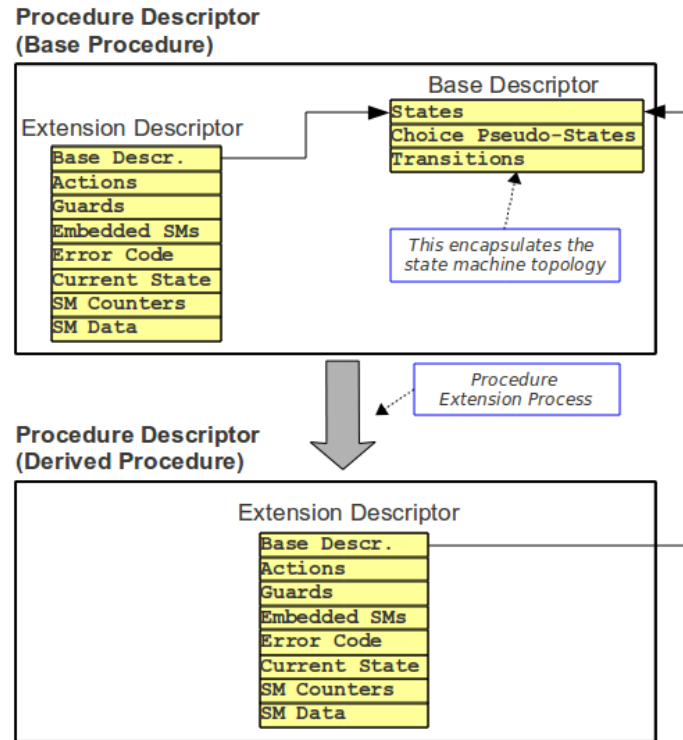


Fig. 7: Extension Mechanism for Procedure Descriptors

The extension mechanism is useful where there is a need to define a large number of procedures which share the same topology (same set of action nodes, of decision nodes, and of control flows) but differ either in their actions or in their guards.

As an example, consider an application which manages a set of hardware devices all of which must be initialized by performing a sequence of elementary operations and suppose that all devices share the same logical sequence of initialization operations (e.g. power-on, switch-on, self-test command, and then, depending on the outcome of the self-test, either a power-off or a command to enter normal operational state). Suppose, however, that the self-tests differ across devices. In this case, it is convenient to proceed as follows:

- A base procedure is defined to model the behaviour which is shared by all devices.
- For each device, a procedure is derived which overrides the action nodes corresponding to the execution of the self-test and the guard which evaluates its outcome.

The pseudo-code of listing 15 offers a concrete example of creation and configuration of a derived procedure. The procedure of Figure 6 acts as *base procedure* and it is extended to create a new procedure (the *derived procedure*). The derived procedure differs from the base procedure in that it has a different action (`incrCnt1By8` instead of `incrCnt1By1`) in its three nodes N1, N2 and N3. Note

that it would not have been possible to override only the action of state N1. The actions of the three nodes N1 to N3 have been defined to be identical and are implemented by the same function `incrCnt1By1` (see configuration examples in the previous section) and can therefore only be overridden together.

In listing 15, the descriptor of the derived procedure is created dynamically by function `FwPrCreateDer`. For users who do not wish to rely on dynamic memory allocation, an alternative approach is available which is illustrated in listing 16. Note that in listing 15 variable `prDescDer` is a pointer to the procedure descriptor whereas in listing 16 it is the procedure descriptor itself.

Use of the derived procedure is done as in the case of non-derived procedures and the examples of section 7.3 remain therefore applicable.

```

1  /* Create the derived procedure (psDesc is the pointer to the PRD
   of the base SM) */
2  FwPrDesc_t prDescDer = FwPrCreateDer(prDesc);
3
4  /* Override Action incrCnt1By1 with action incrCnt1By8 */
5  FwPrOverrideAction(prDescDer, &incrCnt1By1, &incrCnt1By8);

```

Listing 15: Dynamic Creation and Configuration of Derived Procedure

```

1  /* Instantiate the derived PR with 1 actions and 8 guards */
2  FW_PR_INST_DER(prDescDer, 1, 8)
3
4  /* Initialize the derived procedure (prDesc is the pointer to the
   PRD of the base SM) */
5  FwPrInitDer(&prDescDer, prDesc);
6
7  /* Override Action incrCnt1By1 with action incrCnt1By8 */
8  FwPrOverrideAction(&prDescDer, &incrCnt1By1, &incrCnt1By8);

```

Listing 16: Static Creation and Configuration of Derived Procedure

8 RT Container Usage

The basic mode of use of a RT Container in the C1 Implementation is as follows:

- The RT Container Descriptor (RTD) is created
- The RT Container Descriptor is configured
- The RT Container is sent notification requests
- The RT Container is shut down

Examples of creation and configuration of a RT Container can be found in the **FwRtMake*** functions of the **FwRtMakeTest.h** module in the Test Suite. Examples of notifications of RT Containers can be found in the **FwRtTestCases.c** module in the Test Suite.

The pseudo-code examples in this section refer to a simple test container which, every time it receives a notification, writes a message to standard output. The full code for the example is defined in the "RT Container Coding Example 1" (file **FwProfile_RtExample1.c** in the delivery file of the C1 Implementation).

8.1 RT Container Descriptor Creation

A RT Container Descriptor (RTD) is a variable of type **struct FwRtDesc**. A container instance is created as in listing 17. Note that instantiation of the RTD also instantiates the POSIX thread which implements the Activation Thread and the POSIX mutex and POSIX condition variable which support its use.

```
1 /* Instantiate a RT Container Descriptor */
2 struct FwRtDesc rtDesc;
```

Listing 17: Creation of a RT Container Instance

8.2 RT Container Descriptor Configuration

After being created, an RTD must be configured. Configuration is done using the functions defined in header file **FwRtConfig.h**. Configuration is done in steps as follows:

1. The RTD is reset.
2. The attributes of the POSIX objects encapsulated in the RTD are set.
3. The functions implementing the actions of the container's Activation Procedure and Notification Procedure are set.
4. The container data are set.
5. The RTD is initialized.

The reset operation must be done first and the initialization operation must be done last. The other three operations may be done in any order. A container may be configured multiple times but configuration may only be done when the container is in state **rtContUninitialized**, i.e. before it has been initialized or after it has been shut down (see section 5.5).

The RTD is reset through function **FwRtReset**. Execution of this function initializes all fields of the RTD to dummy but valid values (the full list of the initialization values can be found in the Doxygen documentation). Thus, after being reset, the RTD is ready to be initialized (but, after being initialized, will not be able to do anything meaningful). Reset of the test container is done at line 17 in listing 18.

A RT Container uses three POSIX objects: a POSIX thread, a POSIX mutex, and a POSIX condition variable. By default, all three objects are created with the POSIX-defined default values for their attributes. If a user wishes to configure any of these objects with non-default attribute values, he should:

1. Instantiate an attribute object of the appropriate type (for instance, of type `pthread_attr_t` for the POSIX thread attributes).
2. Configure the attribute object as desired using the functions defined by POSIX.
3. Load the configured attribute object into the RTD by means of function **FwRtSetPosixAttr**.

In the test container, default values are used and hence none of the above steps is executed.

By default, the functions implementing the actions of the container's Activation Procedure and Notification Procedure are initialized to dummy functions which do nothing and always return 1. For each such function, the **FwRtConfig.h** module offers a setter and a getter function which can be used to set and read the corresponding procedure function. Thus, for instance, functions **FwRtSetInitializeActivPr** and **FwRtGetInitializeActivPr** can be used to set and get the the function which implements the initialization action for the Activation Procedure.

At a minimum, the user should load a non-trivial Functional Behaviour through function **FwRtSetExecFuncBehaviour**. The function thus loaded defines the behaviour executed by the Activation Thread when it is notified. In the case of the test container considered in this section, function **UserFunctionalBehaviour** in listing 18 represents the user-defined functional behaviour which is executed when the container is notified. This behaviour is encapsulated in a function whose pointer is passed to the container through the call to function **FwRtSetExecFuncBehaviour** in the last line of listing 18.

Other aspects of a container's behaviour (e.g. the behaviour it should execute when it is initialized or when it is shut down) can be defined in a similar way by first defining a function encapsulating the desired behaviour and by then loading a pointer to that function into the container using the appropriate setter function (e.g. the initialization function for the Activation Thread is loaded using function **FwRtSetInitializeActivPr**).

The final step in the configuration process of a RT Container is the definition of the container data (see section 5.6). This is only needed in the case where either data must be passed to the container when it is notified or some of the container functions must exchange data with each other. The container data

are loaded into the container using function `FwRtSetData`. In the case of the test container, no container data are needed and this function is therefore not used.

```

1  /**
2   * Function implementing the user's functional behaviour.
3   * In this example, this function prints a message and returns zero
4   *
5   * @param rtDesc the RT Container descriptor
6   * @return always return zero
7   */
8  FwRtOutcome_t UserFunctionalBehaviour(FwRtDesc_t rtDesc) {
9      static int i = 1;
10     printf("Activation Thread: Notification %i has been received!\n",
11           i);
12     i++;
13     return 0;
14 }
15
16 . . .
17
18 /* Reset the RT Container */
19 FwRtReset(&rtDesc);
20
21 /* Attach functional behaviour to RT Container */
22 FwRtSetExecFuncBehaviour(&rtDesc, &UserFunctionalBehaviour);
23
24 /* Initialize the RT Container */
25 FwRtInit(&rtDesc);

```

Listing 18: Configuration of a RT Container Instance

8.3 RT Container Descriptor Notification

In accordance with the FW Profile, before being able to process notification requests, a RT Container must be started. This is done with the `FwRtStart` function. Applications can check whether a RT Container has already been started with function `FwRtGetContState` which returns the container's state.

After a RT Container has been started, it will respond to notification requests (but note that sending such a request to a container which has not been started is not an error - it simply means that the notification request is ignored). A notification request is sent to a RT container by means of function `FwRtNotify`.

A RT Container is stopped with function `FwRtStop`. After being stopped, the container terminates the Activation Thread and no longer processes notification requests but it may still hold some system resources. If it is desired to ensure that all such resources are released, the container must be shut down by means of function `FwRtShutdown`. This function destroys the POSIX objects used by the container and releases any system resources they may have claimed.

After a RT Container has been stopped, it can be re-started. The start-stop cycle can be executed as many times as desired. After it has been shut down, the container must be re-configured anew before it can again be used.

When a RT Container is stopped (or when it terminates autonomously because

its functional behaviour has terminated execution), its Activation Thread is terminated. A RT Container should only be shut down or re-started after the Activation Thread has terminated. As a convenience, the RT Container provides function `FwRtWaitForTermination` to wait until the Activation Thread has terminated. This function is implemented using POSIX's `pthread_join` system call.

Listing 19 shows a sequence of ten notifications separated by waits of 10 milliseconds for the test container. After the notifications have been sent, the container is stopped and (after its Activation Thread has terminated execution), it is shutdown.

```
1  /* Start the RT Container and send a few notifications to it */
2  FwRtStart(&rtDesc);
3  for (i=0; i<10; i++) {
4      printf("Sending notification %i to container ...\n",i+1);
5      FwRtNotify(&rtDesc);
6      nanosleep(&ten_ms,NULL); /* wait ten ms */
7  }
8
9  /* Stop the RT Container */
10 FwRtStop(&rtDesc);
11
12 /* To ensure orderly shutdown: wait until container thread has
13    terminated */
14 FwRtWaitForTermination(&rtDesc);
15
16 /* Shutdown the RT Container */
17 FwRtShutdown(&rtDesc);
```

Listing 19: Notification of a RT Container Instance

9 Implementation Issues

This section discusses the implementation aspects which have a direct relevance to users of the C1 Implementation.

9.1 Memory Management

The C1 Implementation allocates memory both on the heap and on the stack. The C1 Implementation does not use any global variables.

For the state machine and procedure modules, allocation on the heap (through calls to `malloc`) is done when a new state machine descriptor (SMD) is created using function `FwSmCreate` or when a new procedure descriptor (PRD) is created using function `FwPrCreate`. Both in the case of state machines and procedures, an alternative approach is available to create a state machine instance or a procedure instance without using dynamic memory allocation (see table 6 for the state machines and table 7 for the procedures).

No dynamic memory allocation operations are performed when a state machine or a procedure is configured or executed. Thus, dynamic memory allocation is only used when a state machine or a procedure is created. An application that does not wish to use dynamic memory allocation during real-time operation can instantiate all its state machines and procedures in the initialization part (when real-time constraints are normally not applicable).

The memory allocated on the heap is the memory required to store an SMD or a PRD. The amount of heap memory required by a C1 Implementation is thus proportional to the number of state machines and procedures instantiated by the user.

Operations are provided to release the heap memory allocated when a state machine or a procedure is created (operations `FwSmRelease`, `FwSmReleaseRec` and `FwPrRelease`). Memory is released through calls to `free`.

The RT container module does not directly use dynamic memory allocation (it never calls `malloc`). However, when a RT Container is initialized with `FwRtInit`, its mutex and condition variable and their attribute objects are initialized through calls to the POSIX functions `pthread_*_init`. Depending on how these POSIX functions are implemented, this may involve the allocation of heap memory. Similarly, when the container is started with `FwRtStart`, its Activation Thread is created with a POSIX system call and this, too, may involve allocation of heap memory.

If these POSIX system calls allocate heap memory and if it is desired to avoid dynamic memory allocation during real-time operation, an application should initialize and start all its RT containers in its initialization part.

Any heap allocation which is done by the POSIX functions would be undone when the thread terminates (either autonomously or as a result of a call to `FwRtStop`) and when the container is shut down through a call to `FwRtShutdown`. The latter function destroys the mutex and condition variable objects and their

attribute objects by calling the appropriate `pthread_*_destroy` functions.

Most C1 Implementation functions allocate some memory on the stack but the amount of memory involved is very limited. A precise assessment depends on the characteristics of the compiler but it is expected to consist, at most, of a handful of pointers and variables of primitive type. Note that the amount of stack memory required by the C1 Implementation is independent of both the number and size of the state machines, procedures, and RT containers created by an application.

9.2 Memory Footprint

There are two aspects to the memory footprint of the C1 Implementation: the memory requirements for the code (i.e. the memory requirements for the code generated by compiling the files in the C1 Implementation modules of table 2) and the memory requirements for the state machine, procedure, and RT container descriptors which applications instantiate. These two aspects are considered separately in the following two sub-sections.

9.2.1 Code Memory Requirements

The C1 Implementation is designed to minimize memory footprint. The exact memory requirements for its code depend on the choice of compiler and linker but will typically be of the order of a few kBytes each for the state machine, procedure and RT container modules. As an example, table 8 reports the memory requirements for the files in the state machine module (but the `FwSmAux.h` functions are not included because they are normally not included in an end application), in the procedure module and in the RT container module. The figures in the table have been obtained with the gcc compiler configured to minimize memory occupation. The data in the table were derived from the linker map. They correspond to the memory of type `.text` (i.e. the code segment containing executable instructions) allocated to each module. The measurements were made on release 1.2.0 of the C1 Implementation in the following environment:

- compiler: gcc version 4.6.3 (Ubuntu/Linaro 4.6.3-1ubuntu5)
- target: i686-linux-gnu
- OS: Linux ubuntu 12.04 (32 bits)
- compiler options: `-Os -Wall -c -fmessage-length=0`
- linker options: `-Wl,-Map=memory.map`

Note that not all applications will need all the files in the table. In particular, the functions in `FwSmDCreate.h/FwPrDCreate.h` and `FwSmSCreate.h/FwPrSCreate.h` files will in most cases be alternative and mutually exclusive. Also, all files but the `FwSmCore.h/FwPrCore.h` file could be dropped by using direct creation and configuration of the state machine descriptor or procedure descriptor. This is discussed in section 6.2 (for state machines) and 7.2 (for procedures).

Table 8: Code Memory Footprint for C1 Implementation Modules

Module	Memory Size	Header File
<i>State Machine</i>	1101 bytes	FwSmDCreate.h
<i>State Machine</i>	326 bytes	FwSmSCreate.h
<i>State Machine</i>	1737 bytes	FwSmConfig.h
<i>State Machine</i>	765 bytes	FwSmCore.h
<i>Procedure</i>	342 bytes	FwPrDCreate.h
<i>Procedure</i>	103 bytes	FwPrSCreate.h
<i>Procedure</i>	1413 bytes	FwPrConfig.h
<i>Procedure</i>	389 bytes	FwPrCore.h
<i>RT Container</i>	884 bytes	FwRtConfig.h
<i>RT Container</i>	857 bytes	FwRtCore.h

9.2.2 Descriptor Requirements

Applications using the C1 Implementation must instantiate a state machine descriptor (SMD) for each state machine instance they deploy, a procedure descriptor (PRD) for each procedure instance they deploy, and a RT container descriptor (RTD) for each RT container they deploy. The memory requirement of an SMD will be considered first.

The memory requirement of an SMD is proportional to both the number of states in the state machine and the number of transitions in the state machine. An exact assessment depends on the characteristics of the compiler (in particular on memory alignment constraints). The minimal requirements (based on the assumption of *packed* memory allocation with no gaps to satisfy alignment constraints) can be computed as a function of the following parameters:

- N_{STATES} is the number of states in the state machine
- N_{CPS} is the number of choice pseudo-states in the state machine
- N_{TRANS} is the number of transitions in the state machine
- $N_{ACTIONS}$ is the number of actions in the state machine (including both state and transition actions; if the same action appears several times, it is counted only once)
- N_{GUARDS} is the number of guards in the state machine (if the same guard appears several times, it is counted only once)
- M_{PNT} is the size of a pointer
- M_{S1} is the size of the `FwSmCounterS1_t` type defined in `FwSmConstants.h`
- M_{U2} is the size of the `FwSmCounterU2_t` type defined in `FwSmConstants.h`
- M_{ERR} is the size of the `FwSmErrCode_t` type defined in `FwSmConstants.h`

An SMD is split into two parts: the base descriptor and the extension descriptor (see Section 3.1). Their memory requirements are:

$$M_{BASE-DESC} = M_{PNT} * 3 + M_{S1} * (5 * N_{STATES} + 2 * N_{CPS} + 3 * N_{TRANS}) + (N_{TRANS} * M_{U2})$$

$$M_{EXT-DESC} = M_{PNT} * (4 + N_{ACTIONS} + N_{GUARDS} + N_{STATES})$$

$$+4 * M_{S1} + 2 * M_{U3} + M_{ERR}$$

The memory requirement for a state machine descriptor of a non-derived state machine (i.e. a state machine which is created from scratch as opposed to being derived from some other state machine) is equal to: $(M_{BASE-DESC} + M_{EXT-DESC})$, whereas the memory requirement for a state machine descriptor of a derived state machine is equal to: $M_{EXT-DESC}$.

As an example, table 9 computes the SMD memory footprint in bytes for a medium-sized state machine.

Table 9: SM Descriptor Memory Footprint (Bytes) Example

State Machine Characteristics	Size
Number of States	5
Number of Choice Pseudo-States	3
Number of Transitions	10
Number of Actions	10
Number of Guards	5
Size in bytes of <code>FwSmCounterS1_t</code>	1
Size in bytes of <code>FwSmCounterU2_t</code>	1
Size in bytes of <code>FwSmErrCode_t</code>	1
Size in bytes of Pointer	4
Memory Footprint in bytes of SMD of Non-Derived SM	195
Memory Footprint in bytes of SMD of Derived SM	109

Consider next the case of a procedure descriptor. The memory requirement of a PRD is proportional to both the number of nodes and the number of control flows in a procedure. An exact assessment depends on the characteristics of the compiler but the minimal requirements (based on the assumption of *packed* memory allocation with no gaps to satisfy alignment constraints) can be computed as a function of the following parameters:

- N_{ANODES} is the number of action nodes in the procedure
- N_{DNODES} is the number of decision nodes in the procedure
- N_{CF} is the number of transitions in the procedure
- $N_{ACTIONS}$ is the number of actions in the procedure (if the same action appears several times, it is counted only once)
- N_{GUARDS} is the number of guards in the state machine (if the same guard appears several times, it is counted only once)
- M_{PNT} is the size of a pointer
- M_{S1} is the size of the `FwPrCounterS1_t` type defined in `FwPrConstants.h`
- M_{U2} is the size of the `FwPrCounterU2_t` type defined in `FwPrConstants.h`
- M_{U3} is the size of the `FwPrCounterU3_t` type defined in `FwPrConstants.h`
- M_{ERR} is the size of the `FwPrErrCode_t` type defined in `FwPrConstants.h`

A PRD is split into two parts: the base descriptor and the extension descriptor (see Section 4.1). Their memory requirements are:

$$M_{BASE-DESC} = M_{S1} * (2 * N_{ANODES} + 2 * N_{DNODES} + 2 * N_{CF})$$

$$+3 * M_{S1}$$

$$M_{EXT-DESC} = M_{PNT} * (2 + N_{ACTIONS} + N_{GUARDS}) \\ +4 * M_{S1} + 2 * M_{U3} + M_{ERR}$$

The memory requirement for a PRD of a non-derived procedure (i.e. a procedure which is created from scratch as opposed to being derived from some other procedure) is equal to: $(M_{BASE-DESC} + M_{EXT-DESC})$, whereas the memory requirement for a PRD of a derived procedure is equal to: $M_{EXT-DESC}$.

As an example, table 10 computes the PRD memory footprint for a medium-sized state machine.

Table 10: Procedure Descriptor Memory Footprint (Bytes) Example

Procedure Characteristics	Size
Number of Action Nodes	5
Number of Decision Nodes	3
Number of Control Flows	10
Number of Actions	3
Number of Guards	8
Size in bytes of <code>FwPrCounterS1_t</code>	1
Size in bytes of <code>FwPrCounterU2_t</code>	1
Size in bytes of <code>FwPrCounterU3_t</code>	4
Size in bytes of <code>FwPrErrCode_t</code>	1
Size in bytes of Pointer	4
Memory Footprint of PRD of Non-Derived Procedure	104
Memory Footprint of PRD for Derived Procedure	65

Consider finally RT container descriptors. RTDs have a fixed structure. Hence, their memory footprint is independent of how a container is configured and is a function of the following parameters:

- M_{THREAD} is the memory occupation of an instance of `pthread_t` (a POSIX thread)
- M_{MUTEX} is the memory occupation of an instance of `pthread_mutex_t` (a POSIX mutex)
- M_{COND} is the memory occupation of an instance of `pthread_cond_t` (a POSIX condition variable)
- M_{PNT} is the size of a pointer
- M_{U2} is the size of the `FwRtCounterU2_t` type defined in `FwRtConstants.h`
- M_{INT} is the size of the `int` type
- M_{BOOL} is the size of the `FwRtBool_t` type defined in `FwRtConstants.h`
- M_{STATE} is the size of the `FwRtState_t` type defined in `FwRtConstants.h`

The memory requirement of an RTD is given by:

$$M_{RTD} = M_{THREAD} + M_{MUTEX} + M_{COND} + 12 * M_{PNT} + 2 * M_{BOOL} + \\ M_{U2} + M_{INT} + M_{STATE}$$

Evaluation of this formula depends on the memory occupation of three POSIX data structures. This will normally vary across systems. In the case of the system used in section 9.2.1 for the code footprint measurement, the size of the three POSIX data structures as reported by the `sizeof` operator is: 4 bytes for `pthread_t`, 24 bytes for `pthread_mutex_t`, and 48 bytes for `pthread_cond_t`. Also, the size of the data types from which RTDs are instantiated (`struct FwRtDesc`) is 144 bytes. These figures do not include the memory requirements of additional data structures which may be created on the heap when the POSIX objects are initialized but they serve to give an idea of the typical memory requirement of an RTD instance.

Note finally that users have some control over the memory requirements of a state machine descriptor because they can override the default definition of the following types: `FwSmCounterU1_t`, `FwSmCounterU2_t`, `FwSmCounterU3_t` and `FwSmCounterS1_t`. The override can be done in file `FwSmConstants.h`. A similar consideration applies to the memory requirement of a procedure and RT container descriptor.

9.3 CPU Requirements

CPU requirements depend on the execution platform and no guarantees about absolute CPU demands can be given by the C1 Implementation. The C1 Implementation is, however, designed to give some guarantees of *scalability*. The term *scalability* is used here to designate the dependency of the CPU resources required by the C1 Implementation on the size and number of state machines and procedures instantiated by a user.

For state machines, the following considerations apply to the CPU required to process a transition request made to a state machine:

- There is no dependency on the number of state machine instances in the application. This is because the C1 Implementation operates on individual state machines in isolation from each other.
- There is no dependency on the number of states in the state machine. This is because for each state, a list of its out-going transitions is maintained.
- The CPU time required to perform a transition is proportional to the number of out-going transitions from a state. This is because, when evaluating a transition request from a certain state, all out-going transitions from that state are evaluated in sequence.

For procedures, the following considerations apply to the CPU required to process an execution request made to a procedure:

- There is no dependency on the number of procedure instances in the application. This is because the C1 Implementation operates on individual procedures in isolation from each other.
- There is no dependency on the number of nodes in the procedure. This is because for each node, a list of its out-going control flows is maintained.
- The CPU time required to process the transition through an action node is fixed and independent of the size of a procedure. This is because an

action node can only have one single outgoing transition.

- The CPU time required to process the transition through a decision node is proportional to the number of outgoing control flows from that decision node. This is because, when evaluating a transition through a decision node, all out-going control flows from that decision node are evaluated in sequence.

RT Containers have fixed structures and therefore no scalability considerations apply to them.

9.4 Concurrency

The operations defined by the State Machine and Procedure modules of the C1 Implementation are passive (they do not use any internal threads) and they operate exclusively on stack variables and on the parameters passed to them by the caller. They are therefore intrinsically thread-safe in the sense that different threads can use them on different descriptors without fear of mutual interference. Hence no locks or other synchronization mechanisms are defined for these modules.

Synchronization mechanisms become necessary when different threads operate on the same state machine or procedure descriptor. In this case, the synchronization mechanisms must be provided by the user application.

The operations defined by the RT Container module are similar to those defined by the State Machine and Procedure modules in the sense that they, too, operate exclusively on stack variables and on caller parameters. However, containers have an internal thread (the Activation Thread) which may compete for access to the RTD with the external user thread which stops/starts a container or which sends notifications to it. Conflicts between the Activation Thread and the external threads which call the container operations are avoided through a combination of built-in exclusion mechanisms and constraints to be enforced by the caller as follows:

- The configuration functions defined in `FwRtConfig.h` are not thread-safe and, additionally, may only be used before the Activation Thread has been created or after it has terminated.
- Functions `RtFwStop`, `RtFwStart`, and `RtFwNotify` are protected by the container mutex which ensures that they are only accessed in mutual exclusion.
- Functions `FwRtStart` and `FwRtShutdown` may only be used before the Activation Thread has been created or after it has terminated.

As a convenience for applications, the RT Container module offers function `FwRtWaitForTermination` which can be used to wait for the termination of the Activation Thread and can therefore help to enforce some of the constraints listed above.

9.5 Recursion

No recursion is used in the Procedure and RT Container modules of the C1 Implementation.

Use of recursion is inevitable in the State Machine Module of the C1 Implementation because recursion is intrinsic to the execution model of state machines: a state machine may (recursively) contain other state machines embedded in one of its states and transition requests are propagated (recursively) to the embedded state machines. The depth of recursion is equal to the depth of nesting of state machines.

The **FwSmMakeTrans** and **FwSmExecute** functions which implement the processing of transition requests for state machines are therefore implemented recursively.

The function **FwSmReleaseRec** which releases the memory allocated to a state machine also uses recursion because it releases the memory allocated to a state machine and to all its embedded state machines. In this case, however, a non-recursive version of the function is available: function **FwSmRelease** only releases the memory allocated to a state machine without releasing the memory allocated to its embedded state machines (which must be released with separate calls to **FwSmRelease**).

Applications which do not wish to use recursion should avoid defining state machines embedded in other state machines.

9.6 Order of Execution

The FW Profile stipulates that if a state or choice pseudo-state in a state machine has two or more out-going transitions with a guard which evaluates to true, the transition which will be taken is the one with the lowest evaluation-order value. For a source with n outgoing transitions, where n is a natural number, the evaluation-order values are unique members of the set $\{1, \dots, n\}$ and are assigned according to the order in which the transitions were added to the state machine. The guards of matching transitions are evaluated in increasing evaluation order, and the first guard which evaluates to TRUE determines the selected transition.

Out-going transitions from a state or choice pseudo-state are therefore evaluated in increasing evaluation order. In the C1 Implementation, the transitions from each source occupy a contiguous segment of the transition array in the state-machine descriptor. The segment is reserved when the source is configured, and the **FwSmAddTrans*** functions in module **FwSmConfig.h** fill its slots in configuration order; the slot position therefore implements the evaluation order.

Similarly, the FW Profile specifies that if a decision node has two or more out-going control flows with a guard which evaluates to true, the control flow which will be taken is the one with the lowest evaluation order. For a decision node with n outgoing control flows, where n is a natural number, the evaluation-order values are unique members of the set $\{1, \dots, n\}$, assigned according to

the order in which the control flows were added to the procedure. The guards are evaluated in increasing evaluation order, and the first guard which evaluates to TRUE determines the selected flow.

Out-going control flows from a decision node are therefore evaluated in increasing evaluation order. In the C1 Implementation, the control flows from each decision node occupy a contiguous segment of the control-flow array in the procedure descriptor. The segment is reserved when the decision node is configured, and the `FwPrAddFlow*` functions in module `FwPrConfig.h` fill its slots in configuration order; the slot position therefore implements the evaluation order.

9.7 User Overridable Types

with one exception, the C1 Implementation does not use any primitive types. Instead, `typedef` are used which are defined in `FwSmConstants.h` (for state machines), `FwPrConstants.h` (for procedures) and `FwRtConstants.h` (for RT containers). Applications can override the definition of any of these types. Normally, the default definitions of these types will be adequate except in cases where an application uses state machines with a very large number of states (more than 128) or procedures with a very large number of nodes (more than 128).

The exception to the rule of always using `typedef` types is the error code in the RT Container module (see section 5.7). The error code is used to store the return value of a POSIX system call. This return value is of type `int`. The error code has therefore been declared to be of the same type.

9.8 Counter Overflow

The only counters which might overflow during procedure or state machine execution (i.e. after all procedures and state machines have been instantiated and created) are the execution counters associated to each state machine and procedure instance.

The state machine and procedure execution counters are defined as instances of types `FwSmCounterU3_t` (for state machines) and `FwPrCounterU3_t` (for procedures). The implementation does not provide any protection against wrap around for these counters. It is the responsibility of each application to verify whether the default definition of `FwSmCounterU3_t` and `FwPrCounterU3_t` is adequate and to update it where necessary (or to accept the possibility of a wrap around in the counter values).

For RT Containers, the only counter which might overflow is the notification counter. This counter is defined as a variable of type `FwRtCounterU2_t`. The implementation does not provide any protection against wrap around for this counter. It is the responsibility of each application to verify whether the default definition of `FwRtCounterU2_t` is adequate and to update it where necessary (or to accept the possibility of a wrap around in the counter value).

A State Machine Model of the FW Profile

The C1 Implementation implements the state machine model of the FW Profile. The FW Profile is defined in reference [1]. For convenience, this section reports an excerpt of reference [1] defining the semantics of state machines.

A.1 Definition of State Machines

A state machine in the FW Profile consists of the following elements:

- One *initial pseudo-state*
- One or more *states*
- One or more *state transitions*
- Zero or more *choice pseudo-states*
- Zero or more *final pseudo-states*
- Two *execution counters*

The *initial pseudo-state* is characterized by one transition which has the initial pseudo-state as its source and has either a state or a choice pseudo-state as its target.

A *state* is characterized by the following elements:

- Zero or more *entry actions*
- Zero or more *do actions*
- Zero or more *exit actions*
- Zero or one *embedded state machine*
- One or more *incoming transitions*
- Zero or more *outgoing transitions*

The state actions represent behaviour which is not decomposed further within the state machine. Actions' behaviour can be defined using natural language or some formalism (e.g. an action language). An embedded state machine is a state machine that is embedded within the state. Embedded state machines are defined in the same way and have the same behaviour as other FW Profile state machines. An incoming transition is a state transition that has the state as its target. An outgoing transition is a state transition that has the state as its source.

A *state transition* is characterized by the following elements:

- One *transition source*
- One *transition target* (or *transition destination*)
- Zero or one *transition trigger* (or *transition command*)
- Zero or one *transition guard*
- Zero or more *transition actions*
- Evaluation order

The transition source and the transition target are either a state or a pseudo-state. The transition trigger is the command that triggers the execution of the transition. A transition guard is a specification that evaluates either to TRUE or to FALSE and has no side effects. Absence of a guard is equivalent to a guard which always evaluates to TRUE. A transition action represents behaviour which is not decomposed further within the state machine. A transition action behaviour can be defined using natural language or some formalism (e.g. an action language). For a source with n outgoing transitions, where n is a natural number, each outgoing transition is assigned an evaluation-order value from the set $\{1, \dots, n\}$ according to the order in which the transition was added to the state machine during the state machine configuration process. Thus, the first transition added has evaluation order 1 and the last transition added to the same source has evaluation order n ; no two outgoing transitions from the same source have the same evaluation-order value. When a transition command is processed, the guards of matching outgoing transitions are evaluated in increasing evaluation order. The selected transition is the one with the lowest evaluation-order value whose guard evaluates to TRUE. In the C1 Implementation, the outgoing transitions of each state or choice pseudo-state occupy a contiguous segment of the transition array in the state-machine descriptor. The segment is reserved when the source is configured, and calls to the appropriate **FwSmAddTrans*** function fill its slots in configuration order; consequently, the slot position implements the evaluation order. Transition commands may carry parameters and may return values. The parameters and return values are not defined further by the FW Profile. They represent parameters that are passed to the actions and values which are returned by them.

A *choice pseudo-state* is characterized by the following elements:

- One or more *incoming transitions*
- One or more *outgoing transitions*

An incoming transition is a state transition that has the choice pseudo-state as its target. An outgoing transition is a state transition that has the choice pseudo-state as its source.

The *final pseudo-state* is characterized by one or more incoming transitions (namely state transitions that have the final pseudo-state as their target). Note that all final pseudo-states are equivalent and therefore it would be legitimate to allow only one single final pseudo-state. The option to have more than one is introduced as a matter of convenience.

The *execution counters* are unsigned integers which are characterized by their value. The first execution counter is called the *State Machine Execution Counter* and the second one is called the *State Execution Counter*.

The following syntactical constraints apply to the definition of the state machine elements:

- **C1:** The same pseudo-state cannot be both source and target for a transition;
- **C2:** The source and target of a transition cannot both be choice pseudo-

states;

- **C3:** The transition that has the initial pseudo-state as source can have neither a guard nor a trigger;
- **C4:** This constraint has been deleted;
- **C5:** Transitions that have a choice pseudo-state as source cannot have a transition trigger;
- **C6:** This constraint has been deleted;
- **C7:** Transitions that have a state as a source must have a transition command;
- **C8:** Transitions can only link states and/or pseudo-states that belong to the same state machine.

The last constraint implies that transitions from an outer state machines to an embedded state machines or vice-versa are not allowed. Note, however, that the same transition command may trigger a transition both in an outer state machine and in one of its embedded state machine.

The following dynamical constraints must be satisfied when a state transition is executed:

- **D1:** Among the outgoing transitions from a choice pseudo-state, at least one must have a guard that evaluates to true.
- **D2:** Transition guards must be free of side effects: their evaluation cannot change the state of the host application.
- **D3:** The state actions (entry, do, and exit actions) and the transition actions and guards must execute in zero logical execution time (i.e. on an infinitely fast processor and in the absence of pre-emption or blocking, they must execute in zero time).

The last constraint implies that the behaviour encapsulated by actions and guards is constrained to be *purely functional*. In practice, this means that actions and guards cannot include time-dependent behaviour or behaviour that depends on synchronization with other flows of executions.

One type of transition command the *Execute* command has a special status in that it triggers the execution of the current state's do-action. The Execute command models the situation (common in embedded control systems) of a cyclical scheduler periodically triggering an application and advancing its execution.

As a matter of terminology, when a state machine is sent the Execute command, the state machine is said to be *executed*.

The execution counters of a state machine count the number of times the state machine has been executed (one counts the number of times the state machine has been executed since it was started and the other counts the number of times the state machine has been executed since its current state was entered). Since state machines will often be executed periodically, the execution counters can serve as proxies for measuring the elapsing of time.

A.2 State Machine Behaviour

Three operations may be performed on a state machine: (a) the state machine may be *started*; (b) the state machine may be *sent a transition command*; or (c) the state machine may be *stopped*. State machines are purely reactive: they wait for one of these three operations to be performed upon them and they only execute some behaviour in response to one of these operations.

A state machine can be either in a defined state or in an undefined state. A state machine is in a defined state from the time it has completed the transition out of its initial pseudo-state to the time it has either completed the transition into one of its final pseudo-states or has been stopped.

When a state machine is in a defined state, it has a *current state*. The current state is one of the states of the state machine.

When a state machine is started, the following behaviour is executed:

1. If the state machine is in a defined state, then no further action is taken.
2. If the state machine is in an undefined state, then its execution counters are reset and the action associated to the transition out of its initial pseudo-state is executed. If several transition actions are present, they are executed in the order in which they are listed.
3. If the destination of the transition out of the initial pseudo-state is a choice pseudo- state, then the guards of the outgoing transitions from the choice pseudo-state are evaluated and the actions associated to the transition with a guard evaluating to true is executed. If several transition actions are present, they are executed in the order in which they are listed.
4. If the destination of the transition out of the initial pseudo-state is a state, then the current state of the state machine is set equal to that state.
5. If the destination of the transition out of the initial pseudo-state is a choice pseudo- state and if the selected transition out of the choice pseudo-state has a state as a target, then the current state of the state machine is set equal to that target state.
6. The entry action of the current state is executed. If several entry actions are present, they are executed in the order in which they are listed.
7. If the current state has an embedded state machine, then the embedded state machine is started.
8. If the destination of the transition out of the initial pseudo-state is a choice pseudo- state and if the selected transition out of the choice pseudo-state has the final pseudo- state as a target, then the state machine remains in an undefined state.

With reference to point 3, it is noted that at least one of the guards on the outgoing transitions from a choice pseudo-state is guaranteed to be true because of constraint D1 in the previous section.

When a state machine is stopped, the following behaviour is executed:

1. If the state machine is in an undefined state, no further action is taken.

2. If the state machine is in a defined state and its current state has an embedded state machine, the embedded state machine is stopped.
3. The exit action of the current state is executed. If several exit actions are present, they are executed in the order in which they are listed.
4. The state machine is set to an undefined state.

When a state machine is started, its previous state is set to an invalid value. Before the current state is changed during execution, its value is copied to the previous state. Consequently, when the initial pseudo-state is left, the previous state remains invalid. For a transition to the same state, the previous and current states are identical. When a state machine is stopped, its previous state and current state are set to invalid values.

The logic of the start and stop commands for state machines is shown in Figure 8 as two activity diagrams.

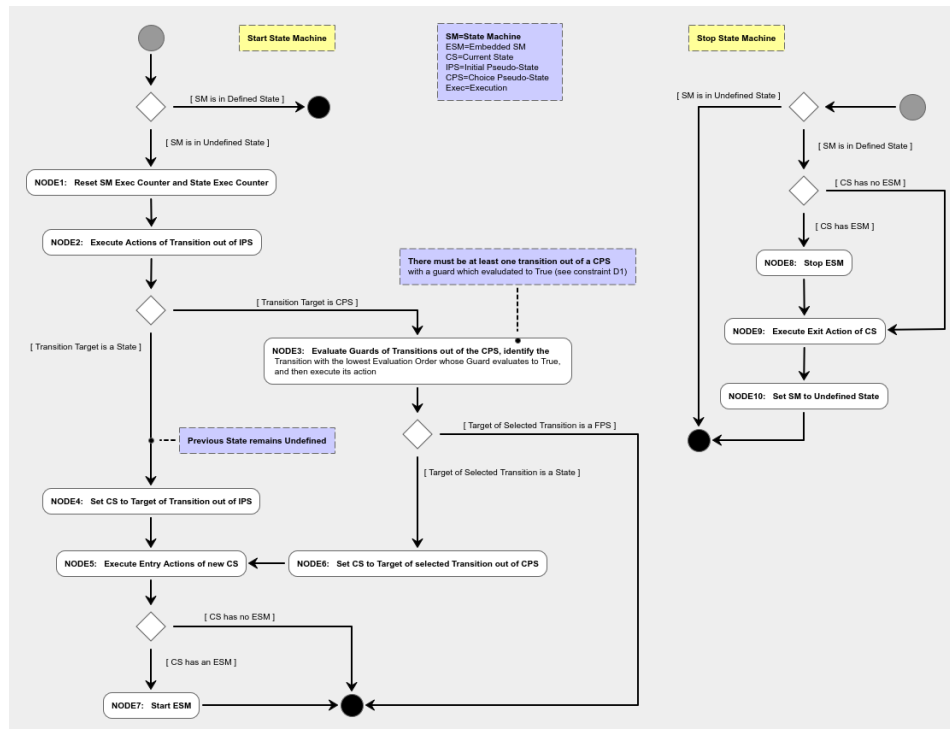


Fig. 8: Logic for the Start and Stop Commands to a State Machine

When a transition command T is sent to a state machine S, then the following behaviour is executed:

1. If S is in an undefined state, then no further action is taken.
2. If T is the Execute command, then the execution counters of the state machine are incremented and the do-action associated to the current state of S is executed. If several do-actions are present, they are executed in the order in which they are listed.

3. If S is in a defined state and the current state of S has an embedded state machine SE, then the transition command T is propagated to SE.
4. If there are no transitions from the current state of S that have T as their trigger, then no further action is taken.
5. If there are one or more transitions from the current state of S that have T as their trigger, then their guards are evaluated in their defined order. The absence of a guard is equivalent to a guard that returns TRUE.
6. When the first transition is found whose guard evaluates to TRUE, then that transition is executed.

The logic that governs the processing of a transition command by a state machine is shown in Figure 9 as an activity diagram. Note that this logic merely describes the circumstances under which a transition within a state machine is executed but it does not define the logic according to which the transition is executed. This is done below (see also Figure 10).

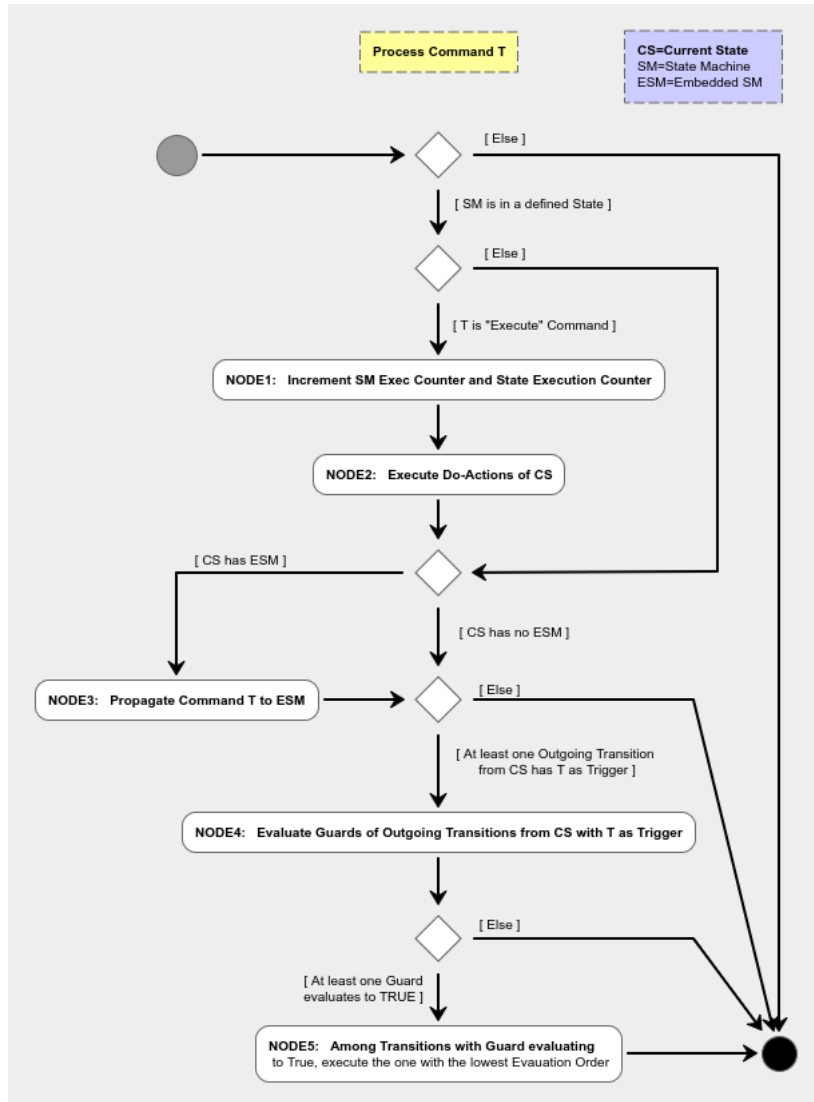


Fig. 9: Logic for Processing Transition Commands by a State Machine

When a transition is executed, then the following behaviour is executed:

1. If the source state of the transition is a state and that state has an embedded state machine, then the embedded state machine is stopped.
2. If the source state of the transition is a state, then the exit action associated to the source state is executed. If several exit actions are present, they are executed in the order in which they are listed.
3. The transition action associated to the transition is executed. If several transition actions are present, they are executed in the order in which they are listed.
4. If the target of the transition is a choice pseudo-state, then the guards of the out-going transitions from the choice pseudo-state are evaluated

in their defined order until one is found that evaluates to true and that transition is executed.

5. If the target of the transition is a final pseudo-state, then the state machine is set to an undefined state and no further action is taken.
6. If the target state of the transition is a state, then the current state of the state machine is updated to be equal to the target state of the transition and the state execution counter is reset.
7. If the target state of the transition is a state, then the entry action of the target state is executed. If several entry actions are present, they are executed in the order in which they are listed.
8. If the target state of the transition is a state and that state has an embedded state machine, then the embedded state machine is started.

With reference to point 4, it is noted that at least one of the guards on the outgoing transitions from a choice pseudo-state is guaranteed to be true because of constraint **D1** in the previous section. The logic according to which a transition is executed is shown as an activity diagram in Figure 10. Note that this logic is called up by the logic shown in the activity diagram of Figure 9.

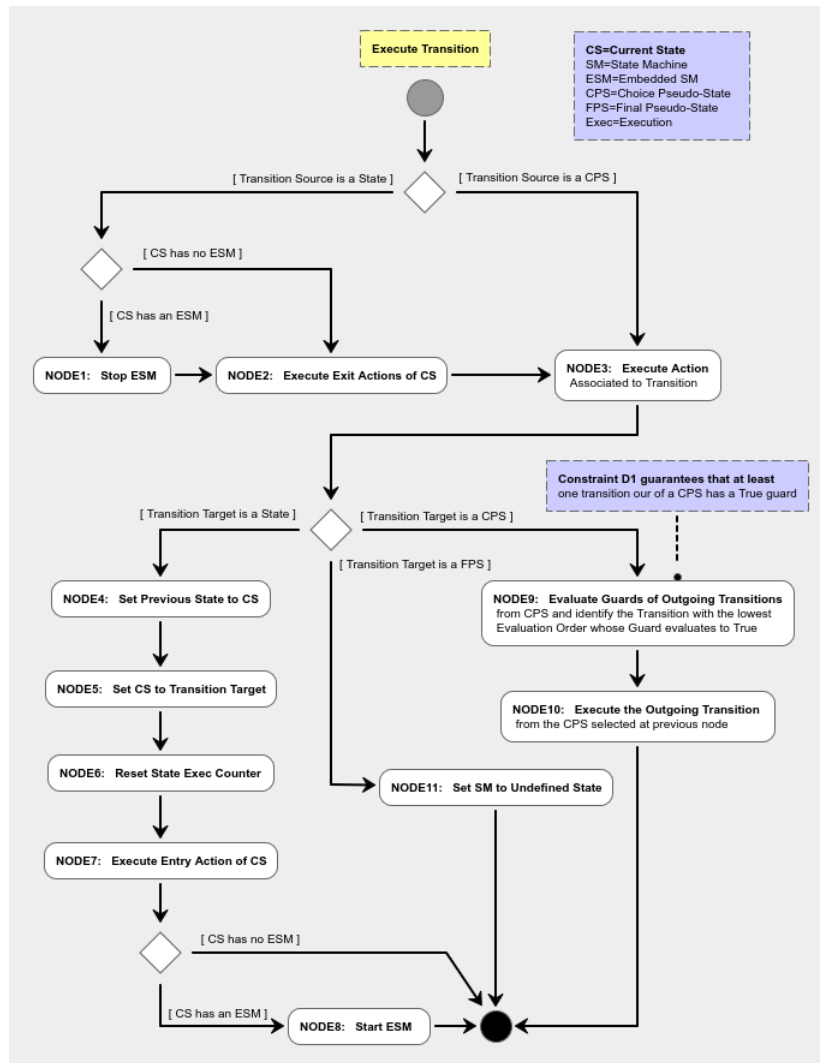


Fig. 10: Logic for Executing Transitions in a State Machine

Transition commands may carry parameters. These parameters may be passed to any of the state or transition actions that are executed as part of the processing of the transition command.

The execution of the various actions associated to the three state machine operations is performed in sequence: an action is executed only when the previous one has completed. Note that, since state and transition actions are constrained to execute in zero logical execution time, the execution of a state machine operation will also execute in zero logical execution time.

Transition commands arrive and are processed in sequence. A new command can only arrive and be processed by a state machine when the previous one has been fully processed. State machines have no queues to buffer incoming transition commands.

The above rule in particular implies that transition commands cannot be nested, namely the processing of a transition command by a state machine cannot result in a new command being sent to the same state machine (nesting rule).

As an example where the nesting rule would be violated, consider the following situation. A first transition command is sent to state machine A that triggers a transition from state A1 to state A2. The entry action of state A2 sends a second transition command to state machine A.

As a second example of violation of the nesting rule, consider a transition command that is sent to state machine A that triggers a transition from state A1 to state A2. The entry action of state A2 sends a new transition command to state machine B. State machine B, as part of its processing of this command, sends a new transition command to state machine A.

Forwarding of transition commands from one state machine A to another state machine B is instead allowed provided that neither of the two state machines is embedded in the other one.

Forwarding of transition commands from an embedded state machine to its embedding state machine or vice-versa is forbidden. This restriction helps to avoid the ambiguities that would arise when, for instance, the entry action of a state in an embedded state machine triggers a transition in the embedding state machine.

A.3 UML 2 Compliance

The state machine model offered by the FW Profile complies with the UML 2 state machine model in the sense that the elements of the state machine concept of the FW Profile and their semantics can be mapped in an obvious way to a subset of the elements of the state machine concept of UML 2 with the following provisos:

- The semantics of choice pseudo-states in the FW Profiles subsumes that of junction pseudo-states in UML2. Thus, in the FW Profile, choice pseudo-states can also be used to join together incoming transition flows.
- The execution counters are specific to the FW Profile. They have been introduced as a substitute for the concept of time (which does not exist in the FW Profile State Machines) in the sense that, if state machines are executed periodically, then the value of their execution counters is proportional to the time elapsed since the state machine was started (State Machine Execution Counter) or since the current state was entered (State Execution Counter).

It should be emphasized that the state machine model proposed by the FW Profile is far more restrictive than that supported by UML 2. This is because the FW Profile uses state machines to model purely functional (non-time-related) behaviour.

B Procedure Model of the FW Profile

The C1 Implementation implements the procedure model of the FW Profile. The FW Profile is defined in reference [1]. For convenience, this section reports an excerpt of reference [1] defining the semantics of procedures.

B.1 Definition of Procedures

A procedure in the FW Profile consists of the following elements:

- One *initial node*
- One or more *actions nodes* (or actions)
- One or more *control flows*
- Zero or more *decision nodes*
- Zero or more *final nodes*
- Two *execution counters*

The *initial node* is characterized by one control flow which has the initial node as its source and has either an action node or a decision node as its target.

An *action node* (or action) is characterized by the following elements:

- One or more incoming control flows
- One outgoing control flow
- The behaviour associated to the action

The incoming control flows are control flows which have the action as its target. The outgoing control flow is a control flow which has the action as its source.

An action represents a single step within a procedure. It encapsulates behaviour that is not decomposed further within the procedure. The action's behaviour can be defined using natural language or some formalism (e.g. an action language).

A *control flow* is characterized by the following elements:

- One source
- One target (or destination)
- Zero or one guards
- Evaluation order

The source and the target are either action nodes or decision nodes. Additionally, the initial node can be the source of a control flow and the final node can be the destination of one or more control flows.

The guard is a specification which evaluates either to TRUE or FALSE and which has no side effects. Absence of a guard is equivalent to a guard which always evaluates to TRUE. For a decision node with n outgoing control flows, where n is a natural number, each outgoing control flow is assigned an evaluation-order value from the set $\{1, \dots, n\}$ according to the order in which the control

flow was added to the procedure during the procedure configuration process. Thus, the first control flow added has evaluation order 1 and the last control flow added to the same decision node has evaluation order n ; no two outgoing control flows from the same decision node have the same evaluation-order value. During procedure execution, the guards are evaluated in increasing evaluation order. The selected control flow is the one with the lowest evaluation-order value whose guard evaluates to TRUE. In the C1 Implementation, the outgoing control flows of each decision node occupy a contiguous segment of the control-flow array in the procedure descriptor. The segment is reserved when the decision node is configured, and calls to the appropriate `FwPrAddFlow*` function fill its slots in configuration order; consequently, the slot position implements the evaluation order.

A *decision node* is characterized by the following elements:

- One or more *incoming control flows*
- Two or more *outgoing control flows*

The incoming control flows are control flows that have the decision node as its target. The outgoing control flow are control flows that have the decision node as their source.

For control flows issuing from a decision node, the pre-defined *Else* guard is available. This guard returns TRUE if and only if all the other guards attached to control flows issuing from the same decision node return FALSE.

The *final node* is characterized by one or more incoming control flows (namely control flows that have the final node as their target). Note that all final nodes are equivalent and therefore it would be legitimate to allow only one single final node. The option to have more than one is introduced as a matter of convenience.

The *execution counters* are unsigned integers which are exclusively characterized by their value. The first execution counter is called the *Procedure Execution Counter* and the second one is called the *Node Execution Counter*.

The following syntactical constraints apply to the definition of the procedure elements:

- C1. The control flows out of a decision node must have a guard.

The following dynamical constraints must be satisfied when a procedure is executed:

- D1. Among the outgoing control flows from a decision node, at least one must have a guard which evaluates to true;
- D2. The evaluation of the guards of a control flow must be free of side-effects;
- D3. The procedure actions and guards must execute in zero logical execution time (i.e. on an infinitely fast processor and in the absence of pre-emption or blocking, they must execute in zero time).

The last constraint implies that the behaviour encapsulated by the actions and by the guards must be purely functional. In practice, this means that actions and guards cannot include time- dependent behaviour or behaviour that depends on synchronization with other flows of executions.

The execution counters of a procedure count the number of times the procedure has been executed (one counts the number of times the procedure has been executed since it was started and the other counts the number of times the procedure has been executed since its current node was entered). Since procedures will often be executed periodically, the execution counters can serve as proxies for measuring the elapsing of time.

B.2 Procedure Behaviour

Four operations may be performed on a procedure: (a) the procedure may be *started*; (b) the procedure may be *executed*; (c) the procedure may be *stopped*; or (d) the procedure may be *run*.

Procedures are purely reactive: they wait for one of these four operations to be performed upon them and they only execute a behaviour in response to one of these operations.

Operations are performed in response to *commands*: the command Start triggers the start operation; the command Execute triggers the execute operation; the command Stop triggers the stop operation; and the command Run triggers the run operation.

A procedure may be in two states: STOPPED or STARTED. Initially, by default, the procedure is in state STOPPED. When the procedure is in state STARTED, it has a *current node*. The current node is either the procedure's initial node or one of its action nodes.

When a procedure is *started*, the following behaviour is executed:

1. If the procedure is in state STARTED, then no further action is performed;
2. If the procedure is in state STOPPED, then it is put in state STARTED, its current node is set equal to its initial node and its execution counters are reset.

When a procedure is *stopped*, the following behaviour is executed:

1. If the procedure is in state STOPPED, then no further action is performed;
2. If the procedure is in state STARTED, then it is put in state STOPPED and its current node is set to an invalid value.

When a procedure starts, its previous node is set to an invalid value. During execution, when control flow enters an action node, the current node is copied to the previous node before being updated to the action node. Consequently, the previous node remains invalid upon leaving the initial node. For a self-transition, the previous and current nodes are identical. Finally, stopping a procedure resets its previous node and current node to an invalid value.

Thus, the Stop and Start commands toggle the state of a procedure and update its current node. This is shown in the state diagram of figure 11.

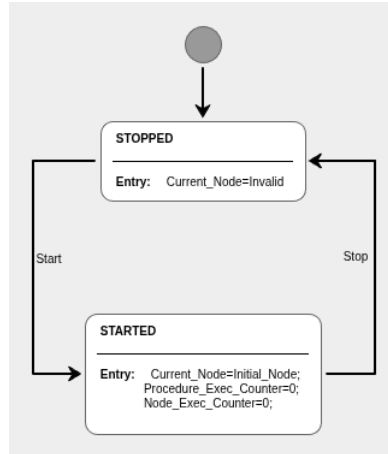


Fig. 11: Procedure Start/Stop Commands

When a procedure is *executed*, the following behaviour is executed:

1. If the procedure is in state STOPPED, then no further action is performed;
2. If the procedure is in state STARTED, then its execution counters are incremented by 1 and the guard attached to the outgoing control flow of the current node is evaluated;
3. If the guard evaluates to FALSE, then no further action is performed;
4. If the guard evaluates to TRUE and the target of the outgoing control flow attached to the current node is an action node T, then: (a) the current node is set equal to T, (b) the node execution counter is reset, (c) the behaviour associated to T is executed, (d) the guard on the out-going control flow of T is evaluated and steps 3 and 4 are (recursively) repeated;
5. If the guard evaluates to TRUE and the target of the outgoing control flow attached to the current node is a decision node, then: (a) the guards of the outgoing control flows attached to the decision node are evaluated in their defined order; (b) if the target of the outgoing control flow whose guard evaluates to TRUE is another decision node, then steps (a) to (d) are performed upon it; (c) if the target of the outgoing control flow whose guard evaluates to TRUE is an action node T, then the current node is set equal to T, the behaviour associated to T is executed, the guard on the out-going control flow of T is evaluated and steps 3 and 4 are (recursively) repeated; (d) if the target of the outgoing control flow whose guard evaluates to TRUE is a final node, the state of the procedure is set to STOPPED and the current node is set equal to an invalid value.
6. If the guard evaluates to TRUE and the target of the outgoing control flow attached to the current node is a final node, then the state of the procedure is set to STOPPED, and the current node is set equal to an invalid value.

Thus, in summary, when a procedure is executed, it tries to traverse the control flow issuing from the current node. If this can be done (i.e. if the guard associated to the control flow evaluates to true), then it advances the execution of the procedure until it finds a guard that evaluates to false or until it finds a final node. Whenever an action node is traversed, its associated behaviour is executed.

The Execute command may carry parameters. These parameters may be passed to any of the actions that are executed as part of the processing of the Execute command.

Note that, at any given time, only one flow of control may be traversing a procedure. This flow of control is advanced every time that the procedure is executed.

The behaviour associated to the execution of a procedure is shown as an activity diagram in figure 12.

Finally, when a procedure is run, the following behaviour is executed:

1. The procedure is started;
2. The procedure is executed;
3. The procedure is stopped.

Thus, the Run operation is defined in terms of the previous three operations. The Run operation may take parameters which are passed to the Execute operation which is performed as part of the Run operation (step 2 above).

The Run operation is only useful for procedures which execute in one single cycle. It is typically used to perform the actions associated to a state in a state machine.

The execution of the various actions associated to the four procedure operations (Start, Execute, Stop, and Run) is performed in sequence: an action is executed only when the previous one has completed. Note that, since actions are constrained to execute in zero logical time, the execution of a procedure operation will also execute in zero logical time.

Requests to perform an operation upon a procedure are executed in sequence. A new request can only be processed by a procedure when the previous one has been fully processed. Procedures have no queues to buffer incoming operation requests.

Note that the procedure operations do not return any values.

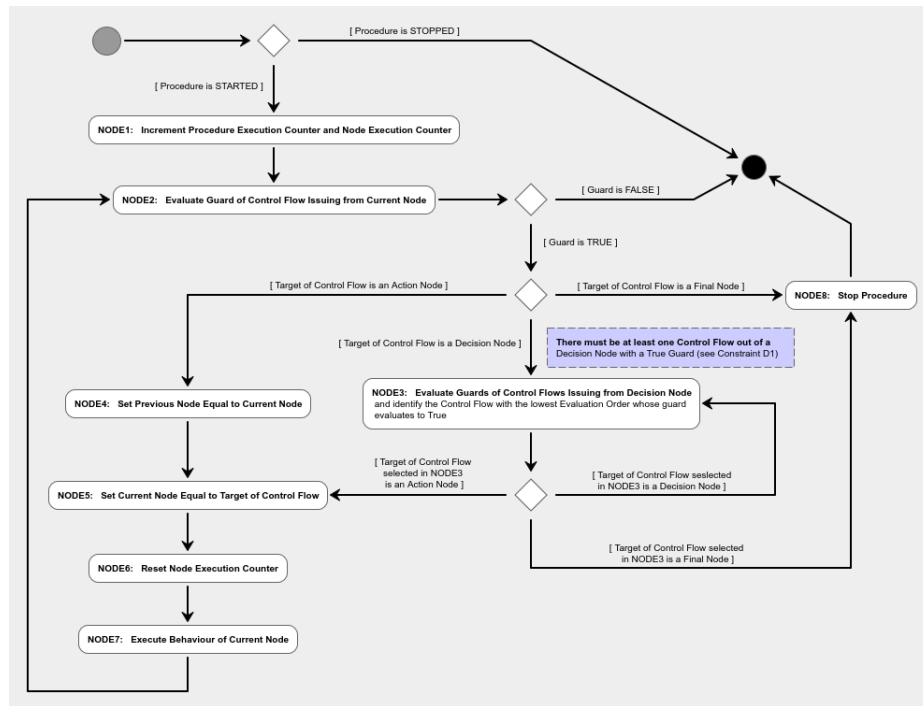


Fig. 12: Procedure Execution Logic

B.3 UML 2 Compliance

The procedure model offered by the FW Profile complies with the UML 2 activity model in the sense that the elements of the procedure concept of the FW Profile and their semantics can be mapped in an obvious way to a subset of the elements of the activity concept of UML 2.

The execution counters are specific to the FW Profile. They have been introduced as a substitute for the concept of time (which does not exist in the FW Profile Procedures) in the sense that, if procedures are executed periodically, then the value of their execution counters is proportional to the time elapsed since the procedure was started (Procedure Execution Counter) or since the current node was entered (Node Execution Counter).

C RT Container Model of the FW Profile

The C1 Implementation implements the RT Container model of the FW Profile. The FW Profile is defined in reference [1]. For convenience, this section reports an excerpt of reference [1] defining the RT Container model of the FW Profile.

C.1 Role of RT Containers

State Machines and Procedures allow all functional aspects of a software application to be modelled. RT Containers complement them by offering a means to capture one aspect of the time-related behaviour of an application.

It is important to stress that full modelling of an application's timing behaviour is beyond the scope of the FW Profile. This is because the FW Profile is aimed at modelling individual applications. Applications normally run on a software/hardware platform which they share with other applications. Timing behaviour is a system-level aspect (it depends, for instance, on the relative priorities of the threads allocated to the various applications in a system) and cannot therefore be fully captured at application level.

RT Containers provide a way to encapsulate the activation logic for a functional behaviour. More specifically, a RT Container can be seen as a representation of a thread that controls the execution of some functional behaviour. The RT Container model defined by the FW Profile allows the conditions under which the thread is released to be specified.

Conceptually, a RT Container can be seen as a software structure that encapsulates some functional code and endows it with certain timing properties. Thus, RT Containers are a means of separating the specification of the timing aspects of an application from its functional aspects.

There is a difference between procedures and state machines on the one hand, and RT Containers on the other hand. All three concepts are offered as means to express the behaviour of a software application but they exist at different levels of abstraction: state machines and procedures constitute a generic modelling language for the functional part of an application; RT Containers allow the timing behaviour of a software application to be modelled but they presuppose the use of certain design patterns for handling the activation of functional code. The RT Container concept is thus less generic than the state machine and procedure concepts.

The design pattern behind the concept of RT Containers is a notification-based model of thread activation where the notification can be either time-triggered or sporadic (event-driven notification).

C.2 Definition of RT Container

A RT Container is defined by the following elements:

- One *Activation Procedure*

- One *Activation Thread*
- One *Notification Procedure*

The *Activation Procedure* is a FW Profile Procedure which executes the functional behaviour encapsulated by the RT Container.

The *Activation Thread* is the thread responsible for executing the Activation Procedure (and hence for executing the functional behaviour encapsulated by the RT Container).

The *Notification Procedure* is a FW Profile Procedure which encapsulates the logic for notifying the Activation Thread.

C.3 RT Container Behaviour

Three operations may be performed on a RT Container: (a) the RT Container may be *started*; (b) the RT Container may be *stopped*; and (c) the RT Container may be *notified*.

A RT Container may be in two states: STOPPED or STARTED. Initially, by default, the container is in state STOPPED. When a RT Container is started, the behaviour shown in the activity diagram in the left-hand side of figure 13 is executed. The Start operation only has an effect if the container is in state STOPPED when the operation causes the Activation and Notification Procedures to be started and executed once and the Activation Thread to be created and released. The Notification and Activation Procedures are started "atomically" in the sense that neither procedure can be executed or stopped before both have been started. Reference to figure 14 shows that the first execution of the Activation and Notification Procedures results in their initialization actions being executed and, in the case of the Activation Procedure, in the first Set-Up Notification action being executed.

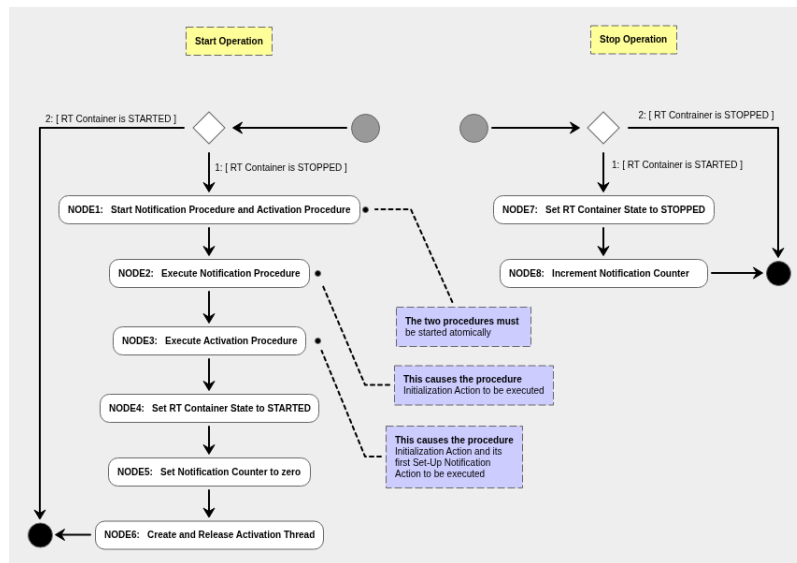


Fig. 13: Start and Stop Operations for RT Containers

When a RT Container is stopped, the behaviour shown in the activity diagram in the right-hand side of figure 13 is executed. The Stop operation only has an effect if the container is in state STARTED when the operation causes the container to be placed in state STOPPED and the Notification Counter to be incremented. The latter results in one last notification being sent to the Activation Thread. This notification is necessary to ensure an orderly termination of the thread and of the Activation and Notification Procedures.

When a RT Container is notified, the following behaviour is executed:

1. If the RT Container is in state STOPPED, then no further action is performed;
2. If the RT Container is in state STARTED, then its Notification Procedure is executed.

The behaviour of the *Activation Thread* is expressed by the following pseudo-code:

```

1 while true do {
2   wait until Notification Counter is greater than 0;
3   decrement Notification Counter;
4   execute Activation Procedure;
5
6   if (Activation Procedure has terminated) then {
7     put RT Container in STOPPED state;
8     execute Notification Procedure;
9     break;
10  }
11
12  if (RT Container is in state STOPPED) then {
13    execute Activation Procedure;
14    execute Notification Procedure;
15    break;
16  }
17 }
```

Listing 20: Pseudo-code of Activation Thread

The thread executes a loop which starts with a check on whether there are any pending notifications (the Notification Counter holds the number of pending notifications). If there is a pending notification (i.e. if the Notification Counter is greater than zero), the thread decrements the Notification Counter and then executes the Activation Procedure (which causes the container's functional behaviour to be executed). The thread terminates when the Activation Procedure has terminated or when the RT container has been stopped. In the former case (Activation Procedure has autonomously terminated), the RT Container is put in the STOPPED state and the Notification Procedure is executed one last time before the thread exits; in the latter case (RT Container has been stopped), both procedures are executed one last time. This last execution is intended to give the procedures a chance to perform their finalization behaviour.

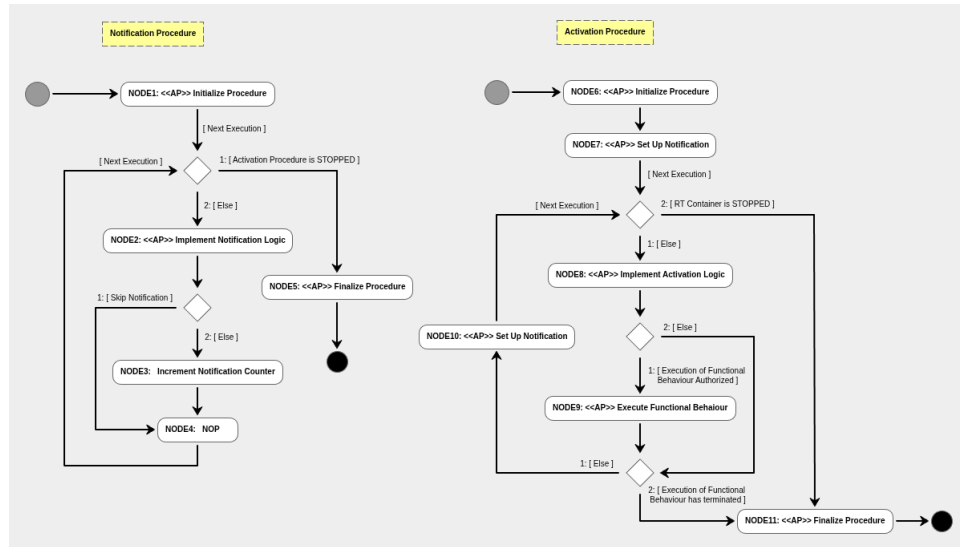


Fig. 14: RT Container Procedures

The behaviour of the *Activation Procedure* and of the *Notification Procedure* is shown in the activity diagrams in Figure 14. The definition of the two procedures makes use of the “adaptation point” stereotype to identify the parts of the container behaviour which are application-specific. Applications are therefore expected to extend the two procedures by inserting their own application-specific behaviour (by contrast, the behaviour of the Activation Thread is invariant and is fully defined at FW Profile level).

When the Activation Procedure is executed for the first time (i.e. after the Activation Thread has been started), it initializes itself and sets up the first notification of the Activation Thread. The form of the notification is application-specific. Typically, the setting up of a notification may consist of one of the following:

1. A request that the Activation Thread be notified at some time in the future;
2. A call-back registration to request to be notified when a certain software condition arises (e.g. a variable changes value, a message arrives, etc);
3. A request to be notified when a hardware interrupt is asserted.

Note that the notification may only need to be set up once when the Activation Procedure is initialized or it may need to be set up at every execution cycle. Note also that the same RT Container may set up different notification requests in the same execution cycle or it may set up notification requests of different kinds at different execution cycles. For this reason, the “Set-Up Notification” in the Activation Procedure is placed both at the beginning of the procedure (to be executed once at initialization time) and inside the loop (to be executed after each execution of the functional behaviour).

When a notification arrives (i.e. when the user of the container executes the Notification Procedure and this increments the Notification Counter), the Ac-

tivation Thread is woken up and it executes the Activation Procedure. The procedure checks whether the RT Container has been stopped. If this is the case, the procedure performs its finalization action and then terminates. Otherwise, the procedure checks whether the functional behaviour should be executed (this is done by the "Implement Activation Logic" action) and, if so, it executes it. Afterwards, the procedure sets up the next notification (if one is needed) and then checks whether the execution of the functional behaviour has been completed. If this is so, the procedure terminates. Otherwise it waits for the next notification.

The procedure initialization and finalization actions are adaptation points which are defined at application level. Similarly, the action to set up the notification for the Activation Thread and to implement the activation logic must also be defined at application level. The latter could, for instance, be used to implement a filter which decides which notifications to process and which ones to ignore.

The Notification Procedure acts as an intermediary between the source of the notification event and the notification trigger to the Activation Thread. Such an intermediary may be useful to: (a) filter notification events, or (b) buffer notification requests so as to allow the Activation Procedure to handle bursts of notifications. With reference to the activity diagram in Figure 14, the filtering and buffering of notification requests is done in the (application-specific) action "Implement Notification Logic".

As already noted, the Notification Procedure runs on a thread that is external to the RT Container: the Notification Procedure is executed by an external thread when the notification event has occurred. Thus, the logic leading to the notification of the Activation Thread is as follows:

1. The Activation Procedure makes a request to be notified when a certain event occurs (this could, for instance, be done by registering with an external component to be notified when a certain condition occurs);
2. When the event occurs, the Notification Procedure is executed by the source of the event;
3. The Notification Procedure evaluates the event and may decide to notify the Activation Thread;
4. The Notification Procedure notifies the Activation Thread by incrementing the Notification Counter;
5. In response to the notification, the Activation Thread executes the Activation Procedure which may execute the functional behaviour encapsulated by the RT Container;
6. The Activation Procedure sets up the next notification request.

This cycle is broken when either the Activation Procedure decides that the execution of the functional behaviour has been completed or when the RT Container is stopped. Either of these events results in the RT Container and its two procedures terminating.

The Notification Procedure may be executed both by the Activation Thread and by an external thread. For this reason, in many cases, it will be necessary

to ensure that it is executed in mutual exclusion.

Note finally that, in this section, the term "event" encompasses both asynchronous occurrences (such as the arrival of hardware interrupts from an external source) or synchronous occurrences (such as periodic signals generated by an operating system).

C.4 RT Container Properties and Usage Constraints

The RT Container logic defined in the previous section guarantees that certain properties (the *RT Container Properties*) are satisfied when the usage of the RT Container complies with certain constraints (the *RT Container Usage Constraints*). The properties are listed in table 11 in rows P-3 to P-7. The usage constraints are listed in the same table in rows C-1 to C-3.

Table 11: RT Container Properties and Usage Constraints

N	RT Container Properties and Usage Constraint
P-3	The Activation Thread shall never deadlock.
P-4	If the RT Container is stopped after the Activation Thread has been released, then, at some later time, the Activation Procedure shall terminate.
P-5	If the Activation Procedure stops or terminates (it enters the STOPPED state), then, at some later time, the RT Container shall be stopped.
P-6	If the Activation Procedure stops or terminates (it enters the STOPPED state), then, at some later time, the Notification Procedure shall terminate.
P-7	Whenever the Activation Procedure is running (it is in state STARTED), then the Notification Procedure shall be running, too (it shall be in state STARTED).
P-8	If notifications cease but the RT Container and the Activation Procedure continue to run, then, at some later time, the Activation Thread shall consume all pending notifications (the Notification Counter will become equal to zero).
C-1	If the RT Container is started and then, at some later time, it is stopped, then it can be re-started only after its Activation and Notification Procedures have terminated execution and after its Activation Thread has terminated (i.e. the user of a RT Container cannot re-start it before it has completed its orderly shutdown)
C-2	The Activation Procedure is started, stopped and executed exclusively by the RT Container (i.e. the user of the container has no access to the Activation Procedure)
C-3	The Notification Procedure is started and stopped exclusively by the RT Container itself (i.e. the user of the RT Container can execute the Notification Procedure through the Notify operation but it cannot start or stop it)

The usage constraints define the conditions for the legal use of a RT Container. If these constraints are satisfied, then the user can assume that the RT Container

will comply with its properties. Note that the container's properties hold under all circumstances, irrespective of the scheduling and notification/triggering policies adopted for the Activation Thread and for the thread controlling the Notification Procedure and irrespective of the way in which the adaptation points in the container's procedure are filled.

Properties P-4 and P-5 guarantee that, if the RT Container is stopped or the Activation Procedure terminates, then the entire container will terminate in the sense that the container itself and its two procedures will all enter the STOPPED state. Property P-8 ensures that, if thread scheduling is fair and the rate at which notifications are generated is compatible with the rate at which they are processed, then no backlog of unprocessed notifications will build up.

Some notifications may instead remain unprocessed if either the Activation Thread autonomously terminates or the RT Container is stopped by the user. Thus, in informal language, the semantics of the Stop operation on the RT Container is not: "Process all pending notifications and then terminate"; but rather: "Discard any pending notifications and then terminate".

Note that the container's procedures can only terminate execution "naturally" (as opposed to being forcefully stopped). This is because the RT Container logic never stops them and usage constraints C-2 and C-3 ensure that they are not stopped by any external agent. This is important because it means that the procedure will always execute their finalization behaviour before terminating.

Constraint C-1 states that a RT Container can only be re-started after it has completed its shutdown. This is a legitimate constraint because properties P-4 and P-6 guarantee that, if the container is stopped, then its two procedures will eventually terminate. This means that the user of a RT Container can always rely on the container completing its shutdown in a finite amount of time.

References

- [1] Alessandro Pasetti, Vaclav Cechticky: *The FW Profile*. PP-DF-COR-00001, Revision 1.3.0, P&P Software GmbH, Switzerland, 2013
- [2] Alessandro Pasetti, Vaclav Cechticky: *The Framework Profile - C1 Implementation User Requirements*. PP-SP-COR-00001, Revision 1.2.2, P&P Software GmbH, Switzerland, 2013